NEW LOWER BOUNDS FOR KISSING NUMBERS IN DIMENSIONS 25 THROUGH 96

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Date: 30 August 2026.

2020 *Mathematics Subject Classification.* Primary 52C17; Secondary 11H31, 11H71, 05B40, 94B25.

Key words and phrases. Kissing number, spherical code, extremal even unimodular lattice, spherical design, Leech lattice, linear programming bound, exact verification.

Abstract. The kissing number $K(n)$ is the largest number of unit vectors in $\mathbb{R}^n$ with pairwise inner products at most $\frac{1}{2}$. We improve the best known lower bound in 48 dimensions between 25 and 96. All but four of these bounds come from the extremal even unimodular lattices Λ_{24}, P_{48} and Γ_{72}, by two constructions: cross-sections of the minimal shell and layered configurations built on it.

For cross-sections we apply Venkov's theorem, that the minimal shell of an extremal even unimodular lattice of dimension divisible by 24 is a spherical 11-design, to monomials in the inner products with several fixed minimal vectors. The resulting linear constraints on the k-point distribution, together with two consequences of the lattice structure, bound the number of minimal vectors orthogonal to all of them. Each bound carries an exact rational dual certificate. Which Gram matrix to fix is settled by Hermite's inequality, and the matrices attaining its bound are exhibited in Γ_{72}: we obtain $K(71) \geq 2\,603\,658\,750$, $K(70) \geq 1\,249\,778\,250$, $K(69) \geq 627\,822\,180$ and $K(68) \geq 361\,275\,480$. In dimensions 46 and 47 the same programme returns the published values 12 309 600 and 23 766 960, both of them exactly, which validates the method.

The layered construction places the minimal shell on an equator of $\mathbb{R}^n \oplus \mathbb{R}^k$ and adjoins lifted copies of part of it. Its level t is a free parameter, bounded below by $1/(2(1-\gamma))$, where γ is the largest inner product allowed between two minimal lines lifted above the same direction. For Λ_{24} and Γ_{72}, $\gamma = \frac{1}{4}$ and the level may be $\frac{2}{3}$, at which one set of lines serves a zero-sum triple of directions; for P_{48}, $\gamma = \frac{1}{3}$ and only an antipodal pair is admissible. Over Λ_{24} the construction gives $K(25) \geq 197\,058$ and $K(27) \geq 200\,540$ and reproduces the published values in dimensions 26 and 28 to 31; rebuilding its caps — as heads in the lens of a minimal vector in dimension 25, and as coset classes over norm-6 vectors placed on every triangle of directions in dimensions 26 and 27 — gives $K(25) \geq 197\,569$, $K(26) \geq 199\,632$ and $K(27) \geq 201\,010$. We obtain $K(38) \geq 591\,612$ and improvements throughout $49 \leq n \leq 61$ and $73 \leq n \leq 95$.

The four remaining bounds are code-theoretic. Evaluated against the current code tables, the chain of Edel–Rains–Sloane gives $K(39) \geq 756\,116$, $K(62) \geq 71\,310\,732$ and $K(63) \geq 138\,419\,844$; at $n = 96$ it overtakes Γ_{72}, giving $K(96) \geq 12\,886\,999\,232$.

Every bound is certified in exact arithmetic by a public verification package; no floating-point computation decides any of them.

1. Introduction

The *kissing number* $K(n)$ is the largest number of non-overlapping unit balls that can touch a central unit ball in $\mathbb{R}^n$. Equivalently, it is the largest size of a set of unit vectors in $\mathbb{R}^n$ whose pairwise inner products are at most $\frac{1}{2}$. Such a set is a *kissing configuration*, or a spherical code of minimum angle 60°.

Exact values are known only for $n = 1, 2, 3, 4, 8, 24$. The cases $n = 3$ [39], $n = 4$ [32], $n = 8$ and $n = 24$ [37, 28] are hard theorems. In every other dimension the known bounds leave an interval whose two ends move independently. This paper concerns the lower end.

1.1. **The published tables.** The state of the art is Cohn's *Table of kissing number bounds* [6, 7], which supersedes the Nebe–Sloane table [35]. Their ranges fix the meaning of "previously known" below. Cohn's table covers dimensions 1–48 and 72; the Nebe–Sloane table covers 1–40, 42, 44, 48, 64, 72, 80 and 128. Above dimension 48 the two together have entries at $n = 64$, 72, 80 and 128 and at no other dimension.

In dimensions 49–63 the best published value therefore comes from the orthogonal direct sum of the minimal shell of P_{48} with a kissing configuration of $\mathbb{R}^{n-48}$ and equals $52\,416\,000+K(n-48)$, where $K(n-48)$ is Cohn's entry in dimension $n-48$, the argument running from 1 to 15. In dimensions 65–71 it is 331 737 984, inherited by monotonicity of K from the dimension-64 entry of Edel–Rains–Sloane [18]. In dimensions 73–96 the same construction over Γ_{72} gives $6\,218\,175\,600+K(n-72)$, which at $n = 96$ is $6\,218\,175\,600+196\,560$, the second summand being the Leech lattice; here $K(n-72)$ is again Cohn's entry, its argument running from 1 to 24. Forty-three of the forty-eight improvements below lie in these ranges.

One dimension inside those three ranges is tabulated: at $n = 80$ Nebe–Sloane record $\Gamma_{72} \perp E_8$, whose kissing number $6\,218\,175\,600 + 240$ is the direct sum above. The rule and the table agree there, and that is the only place they meet.

A table entry is not the only thing a bound above dimension 48 has to beat. The construction of Edel–Rains–Sloane is generic in n, and evaluated in the dimension itself it can exceed the value inherited from a smaller one: in dimensions 65–71 it does, by a little, giving 331 771 800 at $n = 68$ (Remark 4.13). The "previously" column of Table 4 is the tabular floor defined here; §8.2 measures every bound of this paper against the constructive one, which is the stronger test and the one on which the dimension-68 bound is tightest.

1.2. **Cross-sections and layers.** Both constructions start from the minimal shell of one of the three extremal even unimodular lattices

$$\begin{array}{llll} \Lambda_{24} & n=24, & \mu=4, & N=196\,560, \\ P_{48} & n=48, & \mu=6, & N=52\,416\,000, \\ \Gamma_{72} & n=72, & \mu=8, & N=6\,218\,175\,600, \end{array}$$

where μ is the minimum and N the number of minimal vectors. The minimal shell of each, rescaled to the unit sphere, is a kissing configuration; Λ_{24} attains $K(24) = 196\,560$.

The first construction takes cross-sections. The minimal vectors orthogonal to k fixed minimal vectors span a subspace of dimension at most $n - k$ and have pairwise inner products at most $\frac{1}{2}$; they therefore form a kissing configuration in $\mathbb{R}^{n-k}$. The problem is to count them, for which Venkov's theorem provides the constraints (§4).

The second construction places the shell on an equator of $\mathbb{R}^n \oplus \mathbb{R}^k$ and adjoins copies of part of it, lifted out of the equatorial subspace above a small set of directions in $\mathbb{R}^k$ (§5). Every published record in dimensions 25 through 31 belongs to this family.

A cross-section loses points and gives a bound below dimension n; the layered construction gains points and gives one above it. The same quantity limits both: how many minimal lines of L can be pairwise at small inner product.

1.3. **Results.** Table 1 groups the improvements by construction. Table 4 in §7 gives one row per dimension.

TABLE 1. The forty-eight improvements, grouped by construction; # is the number of dimensions in a row. The gain is $K_{\text{here}} - K_{\text{prev}}$ and the factor their ratio, computed dimension by dimension, with both extremes attained at an end of the dimension range, to the two decimals shown. "Previously" is [6] where that table has an entry and otherwise the value of §1.1. Dimensions 44 to 47 are absent, for the reasons given with Table 4, which gives every dimension.

construction	dimensions	#	largest gain	factor	where
Γ_{72} cross-sections	68–71	4	+2 271 920 766	1.09–7.85	§4.8
layers over Λ_{24}	25–27, 38	4	+24 960	1.00–1.04	§5.4–5.7
layers over P_{48}	49–61	13	+7 481 540	1.00–1.14	§5.8
the ERS chain	62, 63	2	+86 001 280	1.36–2.64	§6.2
the ERS chain at 96	96	1	+6 668 627 072	2.07–2.07	§6.3
layers over Γ_{72}	73–95	23	+254 796 296	1.00–1.04	§5.8
constant-weight codes	39	1	+128	1.00–1.00	§6.1
total		48			

In Table 1 the factor is largest where the published table has no entry at all. Three rows exceed 2, and in none of them is the gain a new construction: the cross-sections count a classical configuration, and dimensions 62, 63 and 96 evaluate a construction of 1998 against the code tables of 2026. The rows with factors near 1 span the most dimensions, 41 of the 48, though the two largest absolute gains lie in rows whose factor exceeds 2.

1.4. **Attribution.** *Dimensions* 46 *and* 47. The value 23 766 960 is due to Boyvalenkov–Cherkashin [3]: it is equation (3) of the preprint `arXiv:2312.05121`, whose surrounding paragraph observes that those A_0 vectors form a kissing configuration in $\mathbb{R}^{47}$. The value 12 309 600 is Table 3 of Sun–Wang [40] (`arXiv:2607.20359v3`), obtained by enumerating the 52 416 000 minimal vectors of P_{48}; the degree-3 Siegel theta tables of Ozeki [38] contain the ingredients a decade earlier. We derive both in §4.7

as a check on the cross-section programme, but claim neither. Cohn's table [6] records 5 318 060 in dimension 46 and 9 741 412 in dimension 47.

The forty-eight listed below. We have found no earlier claim in dimensions 25–27, 38, 39, 49–63, 68–71 or 73–96. The one-point distribution of the minimal shell of Γ_{72} follows from Venkov's theorem in a few lines; the bounds in dimensions 70 and 71 may therefore already be known to specialists, though we have found no record of them. Dimensions 62, 63 and 96 carry no new mathematics either. They evaluate the construction of [18] at three lengths where we find no record of its having been evaluated: its seven worked examples are dimensions 32, 36, 40, 44, 64, 80 and 128. All three improve whenever the code tables do. The range "32 to 128" of that paper's abstract is the span of those seven examples; it tabulates no value by dimension, and dimension 96 does not appear in it. Its own totals depend on the code tables in the same way, and it says in closing that it does not expect them to stand as those tables improve.

1.5. **Organisation.** §2 places this work in the literature. §3 fixes notation and states the two classical inputs. §4 develops the cross-section method and proves the bounds in dimensions 68 through 71. §5 analyses the layered construction and proves the remaining bounds, apart from dimensions 39, 62, 63 and 96, which are code-theoretic and are treated in §6. §7 collects the full table. §8 records the constructions we find to be exhausted, the margin of the bounds above dimension 48, one open question and the verification.

A public software package [25] verifies every bound below, to the standards set out in §8.5; no floating-point computation decides any of them.

2. Related work

2.1. **Upper bounds.** The results below concern lower bounds. The upper end is dominated by linear and semidefinite programming. The method of Delsarte [15], in the spherical form of Delsarte, Goethals and Seidel [16], yields the classical bounds of Odlyzko and Sloane [37] and of Levenshtein [28], which determine $K(8)$ and $K(24)$. Semidefinite relaxations strengthen it. Bachoc and Vallentin [1] introduced the three-point bound; Mittelmann and Vallentin [31] computed it to high accuracy; Machado and de Oliveira Filho [30] reduced it by polynomial symmetry. The clustered low-rank solvers of Leijenhorst and de Laat [27] produce most of the upper-bound column of [6] from dimension 10 upwards and establish the optimality of the D_4 root system [14]. The gap remains wide in the dimensions treated here: in dimension 47 the best known construction gives 23 766 960 (§4.7) against an upper bound of 621 658 419.

2.2. **Sphere packing.** The kissing number problem is the local form of the sphere packing problem. The two have shared techniques since Cohn and Elkies [8] introduced the linear programming bound for packing density. Viazovska [43] settled the packing problem in dimension 8, and Cohn, Kumar, Miller, Radchenko and Viazovska [10] in dimension 24; the same authors later proved universal optimality of E_8 and Λ_{24} [11]. Each of these proofs exhibits a modular form making the linear programme tight. Dimensions 8 and 24 are exactly the dimensions in which the kissing number was already known. No such form is available in the dimensions considered here.

2.3. **Constructions from codes.** The entries of [6] for $32 \le n \le 47$ come from the Edel–Rains–Sloane construction [18], with the single exception of dimension 43. That construction assembles a 60° code from a chain of binary constant-weight codes and is therefore only as strong as the code tables it is evaluated with [4]. Write $A(n, d)$ for the largest size of a binary code of length n with minimum Hamming distance d, and $A(n, d, w)$ for the largest size of such a code all of whose words have weight w. Lower down, Zinoviev and Ericson [45] improved a range of small dimensions by an extension scheme built on Steiner systems. Of their values only dimension 13 still stands; the work of §2.4 and §2.5 has overtaken the rest. Improvements to the underlying code tables propagate directly. Echols [17] reports 124 new constant-weight codes together with the kissing numbers they give in dimensions 32, 33, 34 and 37; §6.1 is a further instance. Sun and Wang [40] obtain dimensions 39, 43 and 45 by an antipode construction applied to cross-sections of P_{48}. The same work recomputes the kissing numbers of twelve Conway–Sloane cross-sections, including the two recorded in §1.4.

2.4. **Algebraic constructions.** Ganzhinov [19] constructs highly symmetric line systems from twisted spherical functions of finite groups and obtains kissing configurations of 510, 592 and 1932 points in dimensions 10, 11 and 14. The dimension-14 configuration provides the directions used in §5.7. Cohn and Li [12] improve dimensions 17 through 21 by reversing the parity of the sign count in a Steiner-system construction, which admits further vectors indexed by a subcode of a punctured Golay code. Ho [22] improves their dimension-19 value to 11 948 by exhibiting a nonlinear subcode of that kind. §8.1 analyses both. In dimensions 25 through 31 the predecessor is Kallal, Kan and Wang [23], whose configurations belong to the family of §5 and were superseded by those of Ma et al. (§2.5).

2.5. **Numerical and machine-assisted search.** A separate line of work obtains configurations by numerical optimisation, and several of the values improved on below come from it.

Ma et al. [29] apply game-theoretic reinforcement learning and hold dimensions 25 through 31. Their configurations are the input to §5.4 and §5.5, which construct no new spherical code. Takhanov, Assylbekov and Yun [41] obtain $K(12) \ge 841$ by energy minimisation initialised from an analysis of the structure common to the known 840-point arrangements.

Evolutionary search over programs driven by large language models has improved a single dimension. AlphaEvolve [36] raised $K(11)$ from 592 to 593; agents on the EinsteinArena platform [2] later raised it to 604. Georgiev, Gómez-Serrano, Tao and Wagner [20] applied AlphaEvolve to 67 problems and examined the kissing number in dimensions 1 through 11. They recover the known lower bound in each dimension and report 593 in dimension 11, adding no further improvement; dimension 12 was not attempted. GigaEvo [24] attempted dimension 12 with the aim of improving on the lower bound 840 and reached 840.

The same means have been applied to a related problem. In [26], by the present author with Khrulkov and Oseledets, the object of study is the spherical code, or Tammes, problem: for fixed N and d, arrange N points on S^{d-1} so as to minimise the maximum pairwise inner product. That work samples 90 configurations at random from the catalogue of best known spherical codes maintained by Cohn [5], in dimensions 8 through 16. It lowers the maximum inner product for the majority of them, by up to 2.4% in relative terms, and reports no kissing number result; the question was not asked there.

2.6. **Scope of the present work.** Numerical search has been effective where a configuration is small enough to be stored as a list of coordinates and refined numerically; recent work of this kind has concentrated on dimension 11. The configurations treated here contain between 1.98×10^5 and 6.4×10^9 points. A lattice together with a few integer parameters specifies each of them. They lie beyond the reach of a coordinate search. We obtain them from the structure of three extremal lattices.

Forty-three of the forty-eight dimensions below lie in ranges in which the published tables have one entry between them, at $n = 80$ (§1.1). The improvements there measure the extent to which those ranges have not previously been examined.

The methods of this paper yield nothing in dimensions 9 through 23. Every published record in dimensions 9 through 19 is maximal, in the sense that no further point may be added to it, as are the cross-sections Λ_{21}, Λ_{22} and Λ_{23}; and the Cohn–Li construction is at its exact ceiling in dimensions 18 through 21 (§8.1).

3. Preliminaries

Throughout, L is an even unimodular lattice of dimension n with minimum μ, and $\Lambda \subset L$ is its set of minimal vectors, or *minimal shell*, with $N = |\Lambda|$. For $v, w \in \mathbb{R}^n$ we write $\langle v, w\rangle$ for the standard inner product and $\hat{v} = v/|v|$.

We write Λ_k for the laminated lattice of dimension k, following [13, Ch. 6], so that $\Lambda_{24} = \Lambda_{24}$; the unsubscripted Λ always denotes the minimal shell of L.

Every result below is a lower bound for K. Where the symbol $K(k)$ occurs inside a construction, the inequality proved holds with any lower bound for $K(k)$ substituted. The numerical values quoted use the best known one.

Lemma 3.1 (the shell is a kissing configuration)**.** *$\{\hat{v} : v \in \Lambda\}$ is a kissing configuration in $\mathbb{R}^n$; hence $K(n) \geq N$.*

Proof. For distinct $v, w \in \Lambda$ the difference $v - w$ is a non-zero lattice vector, so $|v-w|^2 = 2\mu - 2\langle v, w\rangle \geq \mu$, giving $\langle v, w\rangle \leq \mu/2$ and hence $\langle \hat{v}, \hat{w}\rangle \leq \frac{1}{2}$. □

L is *extremal* if its minimum is the largest permitted by the theory of modular forms, namely $\mu = 2\lfloor n/24\rfloor + 2$.

Proposition 3.2 (extremal theta series)**.** *The theta series of an extremal even unimodular lattice of dimension n is the unique modular form of weight $n/2$ for $\mathrm{SL}_2(\mathbb{Z})$ with constant term 1 whose coefficients of $q^1, \ldots, q^{\lfloor n/24\rfloor}$ vanish. In particular (μ, N) depends only on n:*

n	8	24	48	72
μ	2	4	6	8
N	240	196 560	52 416 000	6 218 175 600.

Four extremal lattices are known in dimension 48: P_{48p} and P_{48q} [13, Ch. 7], together with P_{48m} and P_{48n}. Nebe constructed the fourth of them [34]; the catalogue [35] records all four. One extremal lattice is known in dimension 72, the lattice Γ_{72} of Nebe [33]. Nothing in §4 distinguishes the four, the bounds there being read off the Gram matrix and the one-point distribution, and we write P_{48} for any of them; the searches on explicit coordinates in §5.8 are carried out in P_{48p}.

Proposition 3.3 (Venkov [42])**.** *If L is extremal even unimodular of dimension $n \equiv 0 \pmod{24}$, then $\{\hat{v} : v \in \Lambda\}$ is a spherical 11-design: every polynomial of degree at most 11 has the same average over Λ as over the sphere. For $n = 8$ the shell is a 7-design.*

Integrality bounds the inner products in the shell.

Lemma 3.4. *For $v, w \in \Lambda$ the inner product $\langle v, w\rangle$ is an integer, and $|\langle v, w\rangle| \leq \mu/2$ unless $w = \pm v$, in which case $\langle v, w\rangle = \pm\mu$.*

Proof. Integrality because L is integral. Both $|v - w|^2 = 2\mu - 2\langle v, w\rangle$ and $|v + w|^2 = 2\mu + 2\langle v, w\rangle$ are norms of lattice vectors, so each is either 0 or at least μ. If neither vanishes, the two bounds give $|\langle v, w\rangle| \leq \mu/2$; and $|v \mp w|^2 = 0$ exactly when $w = \pm v$, which gives $\langle v, w\rangle = \pm\mu$. □

Thus for P_{48} the inner products lie in $\{0, \pm1, \pm2, \pm3\} \cup \{\pm6\}$, and for Γ_{72} in $\{0, \pm1, \pm2, \pm3, \pm4\} \cup \{\pm8\}$.

4. Cross-sections of extremal lattices

4.1. Cross-sections and their cells.

Lemma 4.1. *Let $u_1, \ldots, u_k \in \Lambda$ be linearly independent and put*

$$S = \{\, v \in \Lambda : \langle u_i, v\rangle = 0 \text{ for } i = 1, \ldots, k \,\}.$$

Then S lies in a subspace of dimension $n - k$ and $K(n - k) \geq |S|$. We call S the cross-section *of Λ by $u_1, \ldots, u_k$.*

Proof. S lies in $\operatorname{span}(u_1, \ldots, u_k)^\perp$. All its members have norm μ; Lemma 3.1 applies verbatim. $\square$

It remains to count S. By Lemma 3.4 the vector $(\langle u_1, v\rangle, \ldots, \langle u_k, v\rangle)$ attached to $v \in \Lambda$ lies in $\mathcal{I}^k$, where $\mathcal{I} = \{0, \pm 1, \ldots, \pm\mu/2\} \cup \{\pm\mu\}$. We call an element $c = (c_1, \ldots, c_k)$ of $\mathcal{I}^k$ a *cell* and write

$$n[c] \;=\; n[c_1, \ldots, c_k] \;=\; \#\{\, v \in \Lambda : \langle u_i, v\rangle = c_i \text{ for } i = 1, \ldots, k \,\}$$

for the number of minimal vectors it holds, the *count* of c. The notation leaves $u_1, \ldots, u_k$ implicit, one cross-section being under discussion at a time; at $k = 1$ its argument is a single integer. Thus $|S| = n[0, \ldots, 0]$, the counts of the $|\mathcal{I}|^k$ cells summing to N. The family $\big(n[c]\big)_{c \in \mathcal{I}^k}$ is the *k-point distribution* of $u_1, \ldots, u_k$.

4.2. Moments from the design property.

Lemma 4.2. *For every exponent vector $p = (p_1, \ldots, p_k)$ of non-negative integers with $\sum p_i \leq 11$ and $\sum p_i$ even,*

$$\sum_c n[c] \prod_{i=1}^{k} \Big(\frac{c_i}{\mu}\Big)^{p_i} \;=\; \frac{N \cdot \mathbb{E}\big[\prod_i \langle \hat{u}_i, \zeta\rangle^{p_i}\big]}{n(n+2)\cdots(n + \sum p_i - 2)}, \tag{1}$$

where ζ is a standard Gaussian vector in $\mathbb{R}^n$ and the numerator is evaluated from the normalised Gram matrix $\big(\langle \hat{u}_i, \hat{u}_j\rangle\big)$ by Wick's theorem.

Proof. The monomial $x \mapsto \prod_i \langle \hat{u}_i, x\rangle^{p_i}$ has degree $\sum p_i \leq 11$, so by Proposition 3.3 its sum over the rescaled shell equals N times its average over the sphere. Writing $x = \zeta/|\zeta|$ and using the independence of $|\zeta|$ and $\zeta/|\zeta|$ gives the normalisation $\mathbb{E}|\zeta|^{2m} = n(n+2)\cdots(n+2m-2)$. The Gaussian moment is a sum over perfect matchings, each pair contributing a Gram entry. Odd total degree vanishes because $\Lambda = -\Lambda$. Finally $\langle \hat{u}_i, \hat{v}\rangle = \langle u_i, v\rangle/\mu$ since all vectors have norm μ. $\square$

Remark 4.3 (mixed exponents). Imposing (1) only for those p with every p_i even gives a valid but weaker programme, whose value depends on the presentation of the Gram matrix. At $k = 3$ the full family has 161 equations and the even-exponent family 56. Write a 3×3 Gram matrix of minimal vectors by its off-diagonal entries. Write $3A_3$ for three times the Gram matrix of the root lattice A_3. The matrix $G = (3, -3, 0)$ satisfies $U^\mathsf{T} G U = 3A_3$ for an integer matrix U with $\det U = -1$; thus G and $3A_3$ describe the same configuration. With even exponents alone the two presentations give 6 462 480 and 5 194 969; with the full family both give 6 687 320.

4.3. Constraints from the lattice structure.

Proposition 3.3 uses only that Λ is a spherical 11-design. The next two lemmas sharpen the bound by using that Λ is the minimal shell of a lattice.

Lemma 4.4 (translations)**.** *Suppose $w = \sum a_i u_i$ is a lattice vector with $|w|^2 = \mu$, and let $\tau_l = \langle u_l, w\rangle$. If $\langle w, v\rangle = \mu/2$ for $v \in \Lambda$, then $w - v \in \Lambda$ and $v \mapsto w - v$ is an involution. Consequently*

$$n[c] = n[c'], \qquad c'_l = \tau_l - c_l,$$

for every cell c with $\sum_i a_i c_i = \mu/2$. If $c' \notin \mathcal{I}^k$ the constraint becomes $n[c] = 0$.

Proof. $|w - v|^2 = 2\mu - 2\langle w, v\rangle = \mu$, so $w - v$ is minimal, with $\langle u_l, w - v\rangle = \tau_l - c_l$. The hypothesis $\langle w, v\rangle = \sum a_i c_i = \mu/2$ depends only on the cell; the involution therefore maps the cell c onto the cell c'. ☐

Lemma 4.5 (lattice range)**.** *For every lattice vector $w = \sum a_i u_i$ in the span, with $|w|^2 = \nu > 0$, and every $v \in \Lambda$,*

$$|\langle v, w\rangle| \le \nu/2 \qquad \text{unless } v = \pm w \text{ (possible only if } \nu = \mu).$$

A cell violating this for some such w is empty.

Proof. $|v \mp w|^2 = \mu + \nu \mp 2\langle v, w\rangle$ is a lattice norm, hence 0 or at least μ. ☐

Lemma 4.5 is strictly stronger than testing positive semidefiniteness of the Gram matrix of $(u_1, \dots, u_k, v)$: for a $60°$ pair, $u_1 - u_2$ is itself minimal, which forces $|c_1 - c_2| \le \mu/2$ and excludes ten cells that semidefiniteness permits. Those ten cells are the zeros of [38, Lemma 4.2].

Two further exact constraints are available. The first is the antipodal symmetry $n[c] = n[-c]$. The second is the set of one-point marginals, which are Lemma 4.2 at $k = 1$; §4.6 solves them in closed form.

4.4. **The linear programme and its dual certificate.** Call the linear programme of this subsection, built from k fixed vectors, the *k-point programme*. Its equality constraints form the *k-point system*.

For a cell c let $G(c)$ be the Gram matrix of $(u_1, \dots, u_k, v)$ for a vector v of norm μ with $\langle u_i, v\rangle = c_i$ for every i. Its entries are determined by c and by the Gram matrix of $u_1, \dots, u_k$, whether or not such a v lies in Λ.

Collect the constraints as $Ax = b$, $0 \le x \le h$, with x, h and s below all indexed by cells. Set $h_c = 1$ when $\det G(c) = 0$ and $h_c = +\infty$ otherwise. The first case is legitimate: a vector $v \in \Lambda$ counted by such a cell lies in $\operatorname{span}(u_1, \dots, u_k)$ and is determined there by c; at most one such v therefore exists. Let e_0 be the indicator of the all-zero cell. The true distribution is feasible, whence

$$|S| \;\ge\; \min\{\, e_0^{\mathsf T} x : Ax = b,\ 0 \le x \le h \,\}.$$

Lemma 4.6 (weak duality)**.** *For any rational vector y, put $s = e_0 - A^{\mathsf T} y$. If $s_c \ge 0$ for every cell c with $h_c = +\infty$, then*

$$\min\{\, e_0^{\mathsf T} x : Ax = b,\ 0 \le x \le h \,\} \;\ge\; b^{\mathsf T} y - \sum_{c\colon h_c < \infty} h_c \max(0, -s_c).$$

Proof. For feasible x we have $e_0^\mathsf{T} x = y^\mathsf{T} Ax + s^\mathsf{T} x = b^\mathsf{T} y + s^\mathsf{T} x$, and $s^\mathsf{T} x \geq -\sum_{h_c<\infty} h_c \max(0, -s_c)$ because $0 \leq x_c \leq h_c$. □

A floating-point solver is used only to propose a candidate y. That y is rounded to rationals and repaired into exact dual feasibility by decreasing the coordinate belonging to the all-ones moment row, whose coefficients are all positive. The right-hand side of Lemma 4.6 is then evaluated in exact rational arithmetic and rounded to an integer downwards. An imprecise y weakens the bound and cannot invalidate it. Applying the same lemma to $-e_0$ certifies an upper bound on $|S|$, rounded upwards. A certified minimum may therefore fall one below the true optimum, and a certified maximum one above it.

4.5. **Realizability.** Call a symmetric $k \times k$ integer matrix G with every diagonal entry equal to μ *realizable in* L if some $u_1, \ldots, u_k \in \Lambda$ have Gram matrix G. A bound computed for G is vacuous unless G is realizable in L.

Lemma 4.7. *Let $u_1, \ldots, u_{k-1} \in \Lambda$ have Gram matrix G', and let $c \in \mathcal{I}^{k-1}$. Suppose the minimum of $n[c]$ over the $(k-1)$-point programme is strictly positive. Then some $v \in \Lambda$ satisfies $\langle u_i, v\rangle = c_i$ for $i = 1, \ldots, k-1$, and the matrix obtained by bordering G' with the row c and the diagonal entry μ is realizable in L.*

Proof. The true distribution is feasible for the $(k-1)$-point programme; its value at the cell c is therefore at least the minimum, hence positive. Take $u_k = v$. □

Lemma 4.7 is the weaker of the two routes to realizability. A sweep over all Gram matrices produces larger apparent bounds whose bordering cells have minimum 0, about which the lemma says nothing; among three-point Gram matrices in dimension 69 it licenses only the pairwise orthogonal one, worth 149 415 783, below the value 331 737 984 of §1.1. Realizability is, however, a property of the lattice and not of the programme. Where the lattice is available in coordinates a k-tuple with a prescribed Gram matrix can simply be *exhibited*, as it is for Γ_{72}: dimensions 68 and 69 of Table 4 are obtained this way.

Which Gram matrix to exhibit is then guided by a measurement. Across the Gram matrices tried here the bound falls as $\det G$ rises — at $k = 3$ from 527 597 644 at $\det G = 352$ to 149 415 783 at 512. At a fixed determinant the presentation costs nothing: at $k = 3$ the sixteen realizable Gram matrices of determinant 256 are sixteen bases of one scaled A_3, and all sixteen certify 627 822 180. Taking the determinant as low as it will go, $u_1, \ldots, u_k$ generate a sublattice M of rank k with $\min M \geq \mu$ and $\det M = \det G$, so Hermite's inequality gives $\det M \geq (\mu/\gamma_k)^k$, where γ_k is Hermite's constant in rank k, with equality exactly for the densest rank-k lattice. For Γ_{72}, where $\mu = 8$, that is 48 at $k = 2$, 256 at $k = 3$ and 1024 at $k = 4$, attained by A_2, A_3 and D_4. Those are the three Gram matrices used in dimensions 70, 69 and 68;

each is exhibited inside Γ_{72}. Dimension 69 gains 100 224 536 over the Gram $(4, -2, 0)$, whose determinant is 352.

4.6. **The one-point distribution.** At $k = 1$ the system of Lemma 4.2 has $\mu/2 + 1$ unknowns and six equations, one for each even degree up to 10; the shell of E_8 is a 7-design, which gives four equations there. The system is over-determined; the surplus equations serve as a consistency check.

Proposition 4.8. *For $u \in \Lambda$ and $c = 0, \ldots, \mu/2$ let $n[c]$ be the number of $v \in \Lambda$ with $\langle u, v\rangle = c$. Then*

lattice	$n[0]$	$n[1]$	$n[2]$	$n[3]$	$n[4]$
E_8	126	56			
Λ_{24}	93 150	47 104	4 600		
P_{48}	23 766 960	12 608 784	1 678 887	36 848	
Γ_{72}	2 603 658 750	1 512 243 200	280 928 256	13 959 168	127 800

with each non-zero c attained by $n[c]$ vectors of each sign. The surplus equations hold in every case.

The first two rows are the root system E_7 and the laminated lattice Λ_{23}, whose kissing numbers are known independently. The Γ_{72} row and Lemma 4.1 give Theorem 4.11; the P_{48} row recovers a published value (§4.7).

4.7. **Dimensions 46 and 47.** Both values below are in the literature (§1.4). We derive them here to test the programme against a published value.

Proposition 4.9. *The cross-section of P_{48} by one minimal vector has exactly* 23 766 960 *elements, and the cross-section by two minimal vectors at $60°$ has exactly* 12 309 600*. Hence $K(47) \geq$* 23 766 960 *and $K(46) \geq$* 12 309 600*. Both values are known, the first from* [3] *and the second from* [40].

Proof. For $k = 1$, Proposition 4.8 gives $n[0] = 23\,766\,960$ and the system determines it uniquely; Lemma 4.1 finishes.

For $k = 2$ take $\langle u_1, u_2\rangle = 3$, a $60°$ pair. Such a pair exists: the one-point programme has value $n[3] = 36\,848 > 0$ at that cell, so Lemma 4.7 applies. Running the programme of Lemmas 4.2–4.6 on the all-zero cell, first minimising and then maximising, returns the same certified value 12 309 600 in both directions. Since the true size lies between the two, it equals 12 309 600. □

For the orthogonal pair the programme gives an interval,

$$10\,659\,120 \;\leq\; |S| \;\leq\; 10\,697\,520,$$

and no bound depending only on the Gram matrix can do better. In the notation of Remark 4.10, row 20 of [38, Table 3] divided by $a((3, 3, 0))$ equals 458 386 320/43, which is not an integer. That quotient is an exact average over ordered orthogonal pairs, so the size varies from pair to pair. The width of the interval reflects that variation.

Remark 4.10 (modular forms). The dimension-46 value also follows from modular forms, with no linear programming. Write $a(T)$ for the Siegel Fourier coefficient of the theta series of P_{48} at the half-integral form T, which counts the tuples of lattice vectors whose Gram matrix is $2T$; here T is listed as its diagonal entries followed by the off-diagonal entries of $2T$, so that minimal vectors give diagonal 3 and pairwise entries $\langle u_i, u_j\rangle$. The degree-3 coefficient at a 60° pair together with a vector orthogonal to both, over the degree-2 coefficient at a 60° pair, is

$$\frac{a\big((3,3,3,3,0,0)\big)}{a\big((3,3,3)\big)} = \frac{23\,775\,066\,324\,172\,800\,000}{1\,931\,424\,768\,000} = 12\,309\,600$$

exactly, with no remainder; the degree-3 coefficients are [38, Table 3]. Sun–Wang [40] obtain the same value by direct enumeration.

4.8. **Dimensions 68 to 71.**

Theorem 4.11. *$K(71) \geq 2\,603\,658\,750$ and $K(70) \geq 1\,249\,778\,250$. The first is the exact size of that cross-section of Γ_{72}.*

Proof. Since $72 \equiv 0 \pmod{24}$, the shell of Γ_{72} is an 11-design by Proposition 3.3. For a fixed $u \in \Lambda$, Lemma 3.4 allows $\langle u, v\rangle = c$ with $|c| \leq 4$, together with $c = \pm 8$ for $v = \pm u$: five unknowns against six moment equations. The system is over-determined, solves in non-negative integers, and the surplus equation holds (Proposition 4.8). Then $\Gamma_{72} \cap u^{\perp}$ is a 71-dimensional lattice of minimum 8 whose minimal vectors are exactly the $n[0]$ vectors orthogonal to u; Lemma 4.1 gives the first bound.

For $k = 2$ take $\langle u_1, u_2\rangle = 4$. Such a pair exists because $n[4] = 127\,800 > 0$. The two-point programme certifies $1\,249\,778\,250$. Sweeping all five admissible inner products $\langle u_1, u_2\rangle = 0, 1, 2, 3, 4$ gives, in that order, $1\,073\,551\,500$, $1\,089\,616\,500$, $1\,117\,183\,200$, $1\,167\,565\,500$ and $1\,249\,778\,250$; the 60° pair is the best of them. □

Unlike dimension 46, the minimum and maximum do not meet here: the interval is $[1\,249\,778\,250, 1\,250\,506\,441]$, of which only the lower end is claimed.

At $k = 3$ and $k = 4$ the moment system no longer forces the bound and the Gram matrix has to be chosen; §4.5 takes one at the Hermite floor. That choice is guided by a measurement, but the bound does not rest on it: each certificate is valid for the Gram matrix it is computed from, and that matrix is exhibited.

Theorem 4.12. *$K(69) \geq 627\,822\,180$ and $K(68) \geq 361\,275\,480$.*

Proof sketch. For $k = 3$ take the Gram matrix with diagonal 8 and off-diagonal entries $-4, -4, 0$, that of A_3 scaled by 4, of determinant 256. For $k = 4$ take that of D_4 scaled by 4, of determinant 1024. By §4.5 these are the least determinants a triple and a quadruple of minimal vectors of Γ_{72} can have. Both are realizable by exhibition. Four minimal vectors of Γ_{72}

with the second Gram matrix are written out in [25]; the rank-4 sublattice they generate has 24 minimal vectors, among which a triple with the first is displayed. The three-point and four-point programmes then certify 627 822 180 and 361 275 480; Lemma 4.1 finishes. □

What a numerically proposed dual can be repaired into depends on the basis; the programme's optimum does not. Of the 104 quadruples of those 24 minimal vectors whose Gram matrix has determinant 1024, ninety-eight yield a repaired certificate, running from 358 774 871 to 361 275 383, a spread of 0.7%; the solver does not finish the other six. Solving for the dual *vertex* instead removes it. Substituting out the constraints that are identifications — $n[c] = n[-c]$, and the translations that read $n[c] = n[c']$, $n[c] = 0$ or $n[c] = 1$ — is exact and takes the $k = 4$ programme from 3473 rows and 1681 columns to 617 and 520; the active system is then solved in exact rationals and $A^{\mathsf{T}} y \le c$ verified over every column. The optimum is 361 275 480, and the numeric primal attains it, so no dual improves on it.

The chain ends at $k = 4$. The certified values fall by factors 0.480, 0.502 and 0.575 as k runs from 1 to 4; reaching 331 737 984 at $k = 5$ would need a factor 0.92, and the Hermite floor there is 4096. We have not computed $k = 5$, where the cell enumeration is 11^k; the trend is evidence that it would fall short, not a proof. Dimensions 65 through 67 keep their inherited values here, and we claim nothing in them.

Remark 4.13 (the Edel–Rains–Sloane value). No table has an entry in dimensions 65–71, so the value to improve on is 331 737 984, inherited from dimension 64 [18]. The chain of §6 reproduces both of their large published totals:

$$2^{28} + 30828 \cdot 2048 + 10416 \cdot 16 + 128 = 331\,737\,984,$$

$$2^{30} + 143780 \cdot 2048 + 20540 \cdot 16 + 160 = 1\,368\,532\,064.$$

An inherited value is not the value the construction takes in the dimension that inherits it. The difference matters in dimension 68, where the margin is smallest. There the chain is $(64, 16, 4, 1)$, not $(68, \dots)$: the level-zero sign code has length 64, the best available at any length up to 68, while the support codes spend all 68 coordinates. It gives 331 771 800 from $A(68, 16, 16) \ge A(64, 16, 16) \ge 30\,828$ and $A(68, 4, 4) = \binom{68}{3}/4 = 12\,529$. The second of these is exact, since $68 \equiv 2 \pmod 6$ admits a Steiner system $S(3, 4, 68)$. Theorem 4.12 exceeds that by a factor 1.089, the narrowest margin over the chain of any bound here not derived from it; §8.2 measures the others. Evaluated at $n = 70$ and $n = 71$, granting it the dimension-80 constant-weight code sizes it has no claim to at those lengths, the chain gives at most 563 125 646, which Theorem 4.11 exceeds by factors 2.2 and 4.6.

4.9. **Validation against known cross-sections.** An invalid constraint inflates the minimum of the programme and makes the lower bound false.

We therefore ran the programme on every cross-section whose size is known independently and checked that the known size lies inside the interval the programme produces. Table 2 reports the outcome.

TABLE 2. Every independently known cross-section, against the interval the method produces for it. Sources of truth: classical values for the laminated lattices; direct enumeration from coordinates; and the published computations of [3, 38, 40]. The two orthogonal Gram matrices over Λ_{24} admit embeddings with different counts: at $k = 3$ both are given, at $k = 4$ (†) one of several.

lattice	configuration	minimum	**truth**	maximum
E_8	$k = 1$	126	**126**	126
E_8	$k = 2$ orthogonal	60	**60**	60
E_8	$k = 2$ at 60°	72	**72**	72
Λ_{24}	$k = 1$	93 150	**93 150**	93 150
Λ_{24}	$k = 2$ orthogonal	43 164	**43 164**	43 164
Λ_{24}	$k = 2$ at $\cos\frac{1}{4}$	44 550	**44 550**	44 550
Λ_{24}	$k = 2$ at 60°	49 896	**49 896**	49 896
Λ_{24}	$k = 3$ orthogonal	19 530	**19 530**, **19 962**	19 962
Λ_{24}	$k = 3$, $2A_3$	27 720	**27 720**	27 720
Λ_{24}	$k = 4$ orthogonal	7 464	**8 616**†	17 400
Λ_{24}	$k = 4$, $2A_4$	15 540	**15 540**	15 540
Λ_{24}	$k = 4$, $2D_4$	17 400	**17 400**	17 400
P_{48}	$k = 1$	23 766 960	**23 766 960**	23 766 960
P_{48}	$k = 2$ at 60°	12 309 600	**12 309 600**	12 309 600
P_{48}	$k = 3$, $3A_3$	6 687 320	**6 687 648**	6 702 736
P_{48}	$k = 3$, $3D_3$	6 687 320	**6 702 080**	6 702 736
P_{48}	$k = 4$, $3D_4$	4 147 967	**4 153 600**	4 165 632
Γ_{72}	$k = 1$	2 603 658 750	**2 603 658 750**	2 603 658 750

Every known size lies inside its interval.

Dimension 68 is a cross-section at $k = 4$, and on the Leech lattice the truth is available there. The programme is tight at the two Gram matrices of least determinant: for $2D_4$, which attains the Hermite floor 64, and for $2A_4$, of determinant 80, the minimum and the maximum coincide with the count. The orthogonal quadruple, of determinant 256 against D_4's 64, shows that this is a property of the good Gram matrices rather than of the programme; at that Gram matrix no programme reading the Gram matrix alone could be exact, the count not being a function of it.

The size of a cross-section is not a function of the Gram matrix. The rows $3A_3$ and $3D_3$ carry two bases of one lattice — A_3 and D_3 are the same abstract lattice — so the programme returns one interval for both, yet their sizes differ: 6 687 648 and 6 702 080, at very nearly its two ends. Over Λ_{24} the effect can be exhibited outright: in the coordinates in which a minimal vector has shape $(\pm 4^2, 0^{22})$, the triples

$$4e_1{+}4e_2,\quad 4e_3{+}4e_4,\quad 4e_5{+}4e_6 \qquad\text{and}\qquad 4e_1{+}4e_2,\quad 4e_1{-}4e_2,\quad 4e_3{+}4e_4$$

are both pairwise orthogonal with every norm μ, so they share the Gram matrix μI_3, and their cross-sections have 19 530 and 19 962 elements — exactly the two ends of the certified interval. Orthogonal quadruples behave the same way, realising 8 616, 8 760 and 9 192 among others. A programme reading the Gram matrix alone can therefore only bracket such a row, and the truth column records one embedding in each of these two.

The rows with $k = 3$ also recover a theorem of Ozeki. The three-point system for a 60° triple has affine solution space of dimension 1, along $n[0,0,0] + 164\,n[3,3,3] = 6\,732\,912$. Ozeki's range $184 \le n[3,3,3] \le 278$ [38, Theorem 6.3] therefore corresponds to $6\,687\,320 \le n[0,0,0] \le 6\,702\,736$, which is exactly the interval certified in Table 2; the outward rounding of §4.4 costs nothing at either end. Ozeki obtains the range from modular forms; the programme obtains it with an exact rational certificate.

5. The layered construction

5.1. **The construction.** Fix a lattice L as in §3 and an auxiliary dimension k, and work in $\mathbb{R}^n \oplus \mathbb{R}^k$. Choose a level $t \in (0,1)$ and write $a = \sqrt{t}$, $b = \sqrt{1-t}$.

A minimal *line* of L is an antipodal pair $\pm u$ with $u \in \Lambda$. Set

$$\gamma = \frac{\lfloor \mu/3 \rfloor}{\mu}, \tag{2}$$

the largest inner-product threshold available in L that does not exceed $\frac{1}{3}$. Thus $\gamma = \frac{1}{4}$ for Λ_{24} ($|\langle u, u'\rangle| \le 1$ of 4) and for Γ_{72} (≤ 2 of 8), while $\gamma = \frac{1}{3}$ for P_{48} (≤ 2 of 6). Let $\Gamma_\gamma(L)$ be the graph on the minimal lines of L in which two lines are adjacent when the cosine of the angle between them exceeds γ in absolute value.

Let $C_1, \ldots, C_r$ be pairwise disjoint independent sets of $\Gamma_\gamma(L)$, and let $Z_i \subset S^{k-1}$ be a set of unit *directions* attached to C_i. The configuration is

$$\begin{array}{lll} \text{equator} & (\hat v,\, 0) & v \in \Lambda,\ v \ne \pm u \text{ for any } u \in \bigcup_i C_i, \\ \text{caps} & (s\,a\,\hat u,\ b\,z) & s \in \{\pm 1\},\ u \in C_i,\ z \in Z_i, \\ \text{poles} & (0,\, w) & w \in W \subset S^{k-1}. \end{array} \tag{3}$$

Every point is a unit vector. We write $Z = \bigcup_i Z_i$ for the set of all directions. Counting, with $|C_i|$ the number of lines in C_i,

(4)

$$\# \;=\; N - 2\sum_i |C_i| \;+\; 2\sum_i |C_i|\,|Z_i| \;+\; |W| \;=\; N + 2\sum_i |C_i|\big(|Z_i| - 1\big) + |W|.$$

Proposition 5.1. *The configuration* (3) *is a kissing configuration in* $\mathbb{R}^{n+k}$ *if and only if*

$$t \le 1, \tag{5}$$

$$t\gamma + (1-t) \le \tfrac{1}{2}, \tag{6}$$

$$\langle z, z'\rangle \le \frac{\frac{1}{2}-t}{1-t} \quad \textit{for distinct } z, z' \in Z_i, \tag{7}$$

$$\langle z, z'\rangle \le \tfrac{1}{2} \quad \textit{for } z \in Z_i,\ z' \in Z_j,\ i \ne j, \tag{8}$$

$$b\,\langle z, w\rangle \le \tfrac{1}{2} \quad \textit{for } z \in Z,\ w \in W, \tag{9}$$

$$\langle w, w'\rangle \le \tfrac{1}{2} \quad \textit{for distinct } w, w' \in W. \tag{10}$$

Proof. There are nine kinds of pair. Five are immediate. Equator against equator is Lemma 3.1 and equator against pole is 0. Equator against cap gives $s\,a\langle \hat v, \hat u\rangle$; since $v \ne \pm u$ forces $|\langle \hat v, \hat u\rangle| \le \frac{1}{2}$, the worst case is $a/2 \le \frac{1}{2}$, which is (5). Cap against pole gives $b\langle z, w\rangle$, which is (9), and pole against pole is (10).

The remaining four are cap against cap, and we take them by how the two caps share a line. On one line and one direction the two signs give $-t+(1-t) = 1-2t$, which is at most $\frac{1}{2}$ because (6) rearranges to $t \ge 1/(2(1-\gamma)) \ge \frac{1}{2}$. On one line and distinct directions the value is $\pm t + (1-t)\langle z, z'\rangle$, worst at $s = s'$, which is (7). On distinct lines of one C_i it is $\pm t\langle \hat u, \hat u'\rangle + (1-t)\langle z, z'\rangle$ with $|\langle \hat u, \hat u'\rangle| \le \gamma$; at $z = z'$ that is at most $t\gamma + (1-t)$, which is (6), and at $z \ne z'$ it is at most $t\gamma + (1-t)\langle z, z'\rangle \le t\gamma + \frac{1}{2} - t < \frac{1}{2}$ by (7), so this case is implied by the two before it. On lines from distinct C_i and C_j only $|\langle \hat u, \hat u'\rangle| \le \frac{1}{2}$ is available, by Lemma 3.1, and $t/2+(1-t)\langle z, z'\rangle \le \frac{1}{2}$ is (8). $\square$

5.2. **Pairs versus triples.** Constraints (6) and (7) oppose one another. Rearranged, (6) becomes

$$t \ \ge\ \frac{1}{2(1-\gamma)}, \tag{11}$$

so a larger γ forces a larger level. Substituting the smallest admissible t into (7) bounds the inner products among directions attached to one independent set.

Corollary 5.2. *At* $t = 1/(2(1-\gamma))$ *the directions attached to one independent set satisfy* $\langle z, z'\rangle \le \gamma/(2\gamma-1)$. *Hence*

$$\begin{array}{lll} \gamma = \frac{1}{4}\ (\Lambda_{24},\ \Gamma_{72}): & t \ge \frac{2}{3}, & \langle z, z'\rangle \le -\frac{1}{2}, \quad |Z_i| \le 3, \\ \gamma = \frac{1}{3}\ (P_{48}): & t \ge \frac{3}{4}, & \langle z, z'\rangle \le -1, \quad |Z_i| \le 2. \end{array}$$

Proof. Substituting $t = 1/(2(1-\gamma))$ into $(\frac{1}{2}-t)/(1-t)$ gives $\gamma/(2\gamma-1)$. If m unit vectors are pairwise at $\langle z, z'\rangle \le c < 0$ then $0 \le |\sum z_i|^2 \le m + m(m-1)c$, so $m \le 1 - 1/c$. With $c = -\frac{1}{2}$ this is $m \le 3$ and with $c = -1$ it is $m \le 2$. $\square$

Corollary 5.2 forces the ceiling $\frac{1}{3}$ in (2): at a larger threshold the right-hand side of (7) falls below -1 and no independent set carries more than one direction. The ceiling is active for Γ_{72}, where the largest threshold below $\frac{1}{2}$ that Lemma 3.4 leaves available is $\gamma = \frac{3}{8}$, that is $|\langle u, u'\rangle| \leq 3$; the lemma itself permits $|\langle u, u'\rangle| \leq 4$, but $\gamma = \frac{1}{2}$ makes (6) require $t \geq 1$. The threshold $\frac{3}{8}$ forces $t \geq \frac{4}{5}$, at which (7) requires $\langle z, z'\rangle \leq -\frac{3}{2}$, which no two unit vectors satisfy.

Over Λ_{24} and Γ_{72} an independent set may therefore carry a *zero-sum triple* of directions: three unit vectors summing to zero and pairwise at 120°. Over P_{48} it carries at most an antipodal pair. By (4) a triple contributes 4 points per line and a pair 2, doubling the gain; no independent set carries more than three directions.

Constraint (9) behaves differently at the two levels used below. At $t = \frac{3}{4}$ we have $b = \frac{1}{2}$, so (9) holds for every z and w; W may then be any kissing configuration of $\mathbb{R}^k$, giving $|W| = K(k)$. At $t = \frac{2}{3}$ we have $b = 1/\sqrt{3}$ and (9) requires $\langle z, w\rangle \leq \sqrt{3}/2$; every pole then lies at least 30° from every cap direction. A rotated copy of the direction set is automatically a 60° code, so only the 30° separation is at issue, and the caps around Z occupy little of the sphere once k is large: at $k = 23$ a single 30° cap covers 2.3×10^{-8} of S^{22}, so all 93 150 of them together cover at most 2.1×10^{-3} and a Haar-random rotation is expected to lose at most 198 directions; the rotation used below loses 76, leaving 93 074. We therefore take W to be a rotated copy of Z with the points that fail (9) removed. The rotation is made rational by the Cayley transform $Q = (I - S)(I + S)^{-1}$, which is an isometry of the form whenever CS is skew-symmetric, so (9) is checked in exact integers. It must be an isometry of $\mathbb{R}^k$ itself and not merely of the coordinates in which Z is presented, which for some cross-sections are more numerous: Z spans 21 dimensions of the 24 that carry Λ_{21}. Where they differ we take $S = \sum_t \big(z_t (C z_t')^{\mathsf{T}} - z_t' (C z_t)^{\mathsf{T}}\big)$ over pairs $z_t, z_t' \in Z$ with $\langle z_t, z_t'\rangle = 0$, all $2r$ of them pairwise orthogonal. Then S is integral, CS is skew, and S annihilates the orthogonal complement of Z, so Q fixes that complement pointwise and is an isometry of the span. On the plane of z_t and z_t', taken in the integral coordinates where $\langle z_t, z_t\rangle = m$, scaling S by $1/\lambda$ makes Q the rotation through $2\arctan(m/\lambda)$, and its denominator is $\lambda^2 + m^2$ for every plane at once — two to four digits, against the 2^{71} that a basis of the span would force. Where the direction set is realised over a ring in which we have not done this we take $W = \emptyset$.

5.3. **Calibration.** We first check (4) in dimensions where the answer is known. Take $L = \Lambda_{24}$, so that $\gamma = \frac{1}{4}$, and take every C_i to be a maximum known independent set, of 248 lines. Take Z to be the root system of rank k normalised to the sphere, partitioned into zero-sum triples where its size permits and into antipodal pairs otherwise; take $|W| = K(k)$ poles. The level is the smallest admissible one, $t = \frac{2}{3}$, in every column but the first: at $k = 1$ the sphere S^0 contains no triple; §5.4 raises the level to $t = \frac{3}{4}$ there. Where $t = \frac{2}{3}$ we take the poles to satisfy the 30° separation of (9). Such a set of $K(k)$ poles is exhibited here in every column. At $k = 2$ and $k = 4$ the poles are vectors of norm $2m$ in the same root lattice, where $|a \pm w|^2$ being a lattice norm forces the 30° separation and $K(k)$ of them are available. At $k = 3$ they are the axis layer of the dimension-27 configuration of §5.5, verified over all of its pairs. At $k = 5$, $k = 6$ and $k = 7$ they are a rotated copy of the root system, constructed as in §5.8 and checked in exact integers there. So no column of Table 3 assumes a pole set: each is exhibited and verified. Then (4) gives Table 3.

TABLE 3. The total (4) over Λ_{24} with independent sets of 248 lines, against the published values [6]. In the third row $a\times b$ means a parts of b directions each, so that $\sum ab = K(k)$; a part of 3 is a zero-sum triple and a part of 2 an antipodal pair. The columns that disagree are treated in §5.4 and §5.5.

k	1	2	3	4	5	6	7
dimension	25	26	27	28	29	30	31
parts of Z	1×2	2×3	4×3	8×3	12×3+2×2	24×3	42×3
(4)	197 058	198 550	200 540	204 520	209 496	220 440	238 350
published	197 056	198 550	200 044	204 520	209 496	220 440	238 350
	+2	=	+496	=	=	=	=

The formula reproduces five independently obtained record configurations exactly. The columns $k = 1$ and $k = 3$ exceed the published value; §5.4 and §5.5 show that the additional points are available. The column $k = 2$ has no slack in this family; §5.6 leaves the family, and dimension 26 moves with it.

5.4. **Dimension 25: the choice of level.**

Theorem 5.3. *$K(25) \geq 197\,058$.*

Proof. Take $L = \Lambda_{24}$, $k = 1$, one independent set C of 248 lines, and $Z = W = \{+1, -1\} \subset S^0$. The published configurations use $t = \frac{2}{3}$, at which $b = 1/\sqrt{3} = 0.577\ldots > \frac{1}{2}$; then (9) fails for the two poles, which are discarded.

The level $t = \frac{2}{3}$ is not forced here. Constraint (11) bounds it from below and (7) from above: for two antipodal directions the latter requires $-1 \leq (\frac{1}{2} - t)/(1 - t)$, that is $t \leq \frac{3}{4}$. At $k = 1$ the sphere S^0 has two points and admits no triple; the whole range $\frac{2}{3} \leq t \leq \frac{3}{4}$ is therefore admissible.

At $t = \frac{3}{4}$ constraint (6) becomes $\frac{3}{4}\gamma + \frac{1}{4} \leq \frac{1}{2}$, i.e. $\gamma \leq \frac{1}{3}$; and for Λ_{24} the thresholds $\gamma = \frac{1}{4}$ and $\gamma = \frac{1}{3}$ define the same condition, since Lemma 3.4 allows only $\langle u, u' \rangle/\mu \in \{0, \pm\frac{1}{4}, \pm\frac{1}{2}\}$ and no admissible value lies strictly between $\frac{1}{4}$ and $\frac{1}{3}$. The independent set is therefore unchanged at 248 lines. At $t = \frac{3}{4}$ we have $b = \frac{1}{2}$, so (9) holds and both poles are admissible. Finally (4) gives $196\,560 + 2 \cdot 248 \cdot (2 - 1) + 2 = 197\,058$. $\square$

Three constraints meet at $t = \frac{3}{4}$. The bound $t \geq \frac{2}{3}$ keeps the independent set at 248 lines, the bound $t \leq \frac{3}{4}$ keeps the antipodal pair admissible, and the poles require $t \geq \frac{3}{4}$. The argument does not repeat for $k \geq 2$: triples exist there and contribute twice as much as pairs, which makes $t = \frac{2}{3}$ optimal; the poles already attain $K(k)$ at that level.

The gain is $+2$ over any independent set used in its place.

Superseded, 2026-09-07. The dimension-25 entry of the repository no longer rests on this level argument. In joint work in progress with H. Cohn and B. Lindow the caps are rebuilt: 1006 heads x of squared length 3 (norm-4 units) sit in the lens of a minimal vector, each removing exactly its owner from the equator, and it is a theorem that two compatible heads never share a removal, so the count is $196\,560+H+2+|E|$ with $H = 1006$ and one further equator point $-\frac{2}{\sqrt{6}}v$ that is not a lattice vector. That gives $K(25) \geq 197\,569$, verified in exact arithmetic by the package `dim25-lens-heads`; the account is a separate note. The level argument above remains the template's own bound and is kept as such. The construction uses the 248-line independent set of [29] unchanged; one of 249 lines would give 197 060 by the same argument.

5.5. **Dimension 27: partitioning the directions.**

Theorem 5.4. *$K(27) \geq 200\,540$.*

Proof. Take $L = \Lambda_{24}$, $k = 3$ and $t = \frac{2}{3}$. Take Z to be the A_3 root system normalised to the sphere, the 12 vertices of a cuboctahedron and a kissing configuration of $\mathbb{R}^3$, and take W to be a rotated copy of it satisfying the 30° separation of (9), so that $|W| = K(3) = 12$. By Corollary 5.2 each independent set may carry a zero-sum triple. The 12 vertices admit a partition into four zero-sum triples; four independent sets therefore suffice, and (4) gives

$$196\,560 + 2 \cdot 4 \cdot 248 \cdot (3-1) + 12 \;=\; 196\,560 + 3\,968 + 12 \;=\; 200\,540. \qquad \square$$

The choice of direction set is constrained, and not merely convenient. The 12-point kissing configuration of $\mathbb{R}^3$ is not unique; two of them are classical, the cuboctahedron and the icosahedron. Icosahedral vertices meet at $\langle z, z' \rangle \in \{\pm 1, \pm 1/\sqrt{5}\}$, so no two of them that are not antipodal reach the $-\frac{1}{2}$ that Corollary 5.2 demands: the icosahedron admits no zero-sum triple at all, and over it the construction would be reduced to six antipodal pairs and 199 548 points. The cuboctahedron has 24 pairs at $-\frac{1}{2}$ and admits exactly two partitions into four triples.

The published configuration [29] groups its twelve directions into parts of sizes $3, 3, 2, 2, 2, 2$. That partition uses five independent sets and deletes $5 \times 496 = 2\,480$ vectors from the equator; four triples use four and delete 1 984, returning 496 vectors. The construction adds no new spherical code: the Leech vectors, the twelve directions, the twelve poles and four of the five independent sets are those of the published configuration; only the assignment of independent sets to directions differs. A script in [25] regenerates the configuration from the dimension-27 block of [7].

The family admits nothing further in this dimension. By (4) the gain is $2\sum_i |C_i|(|Z_i|-1)$ with $|Z_i| \le 3$ and $\sum_i |Z_i| \le |Z| = 12$, so $\sum_i(|Z_i|-1)$ is at most $\lfloor 2\cdot 12/3\rfloor = 8$, which four triples attain.

Superseded, 2026-09-08. The dimension-27 entry of the repository no longer rests on Theorem 5.4; see §5.6. The theorem and the partition argument remain the template's own bound, and the twelve directions they fix are the ones the new configuration uses.

5.6. **Dimensions 26 and 27: the coset triangle on every triangle of directions.**

Theorem 5.5. *$K(26) \ge 199\,632$ and $K(27) \ge 201\,010$.*

Both are joint work in progress with H. Cohn and B. Lindow, and the account is the separate note of §5.4; we record the shape and the verification. Write a point of $\mathbb{R}^{24+k}$ as (x, y) with $y \in \mathbb{R}^k$, normalise every point to squared length 4 so that compatibility is $\langle P, Q\rangle \le 2$, and put the $K(k)$ axis points $(0, a)$, $|a| = 2$, on a hexagon ($k = 2$) or a cuboctahedron ($k = 3$). A cap point (x, y) then has $|y| \le \sec\theta$ for the covering radius θ of the axis set; for the hexagon $\theta = 30^\circ$ forces $|x|^2 \ge 8/3$ with equality forcing y onto the six edge midpoints, a head at $|x|^2 = 8/3$ carries a zero-sum triangle of directions, and two cap points there are compatible iff $\langle x, x'\rangle \le 2 - \frac{4}{3}\cos\angle(y, y')$: $\frac{2}{3}$ on a common direction, $\frac{4}{3}$ at 60°, no condition at 120° or 180°. So the layer is one head set per triangle of directions — two for the hexagon, four for the cuboctahedron, whose twelve vertices are four zero-sum triangles every two of which have a pair at 60° — each a code of cosine at most $\frac{1}{4}$ on the sphere $|x|^2 = 8/3$, cross-constrained at cosine $\frac{1}{2}$, and the count is $196\,560 + K(k) + 2(P - F) + 3F$ for P heads of which F remove nothing.

Every head is $y/3$ with $y = 3u+v$ integral, u a minimal vector (the owner, which the head removes) and v a Leech vector of norm 6 with $\langle u, v\rangle = -3$, or a free head $y = \pm 2v$; a head removes exactly its owner, and two heads on one owner have $\langle y, y'\rangle \ge 16$, above every threshold, so removals are not shared. Three such v with $v_1 + v_2 + v_3 = 0$ carry 1656 heads over three disjoint blocks $\Sigma(v_i) = \{u : \langle u, v_i\rangle = -3\}$ of 552, and on one side the exact optimum is 762 of them, against 552 for a single v, which is Cohn's construction. In dimension 26 the same triangle serves both sides: with $\iota(u) = -v_i - u$, a fixed-point-free involution of each block, $\langle 3u + v_i, 3\iota(u') + v_i\rangle = 15 - 9\langle u, u'\rangle$, which respects the cross threshold 12 exactly when $\langle u, u'\rangle \ge 1$; so a side S and its image $\iota(S)$ fit on the two triangles of the hexagon whenever the owners of S are pairwise at inner product 1 or 2 inside each block, and an optimal S with that property exists. The shipped configuration has 1524 class heads and 6 free heads: $196\,560 + 6 + 2\cdot 1524 + 18 = 199\,632$. In dimension 27 the axis is the cuboctahedron rotated by 45°

about a coordinate axis, whose largest cosine to a direction is $(2+\sqrt{2})/4$ and whose inner product with a cap point is at most $(2+\sqrt{2})/\sqrt{3} < 2$; two triangles of leans with cross Gram matrix the circulant of $(3, 0, -3)$ carry 2210 class heads over the four sides and 6 free heads: $196\,560 + 12 + 2 \cdot 2210 + 18 = 201\,010$. The package `dim26-27-iota-triangles` stores the heads as integer vectors with the index of their triangle, rebuilds the Leech vectors from the Golay code, decides every head–equator and head–head pair by integer comparison against the thresholds $144 - 96c$ (c the exact largest cosine between two triangles of directions, in the scaling with minimum 32) and the axis inequalities in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and rejects a head moved, duplicated or displaced.

5.7. **Dimension 38: large codimension.** For $k \le 7$ the number of parts of Z binds in (4), and a part is worth most when it is a triple: the number of triples available, $\lfloor K(k)/3 \rfloor$, is $0, 2, 4, 8, 13, 24, 42$, so at most 42×248 lines carry caps; one therefore takes the independent sets as large as possible. At larger k the parts of Z outnumber the disjoint maximum independent sets available, and the binding question becomes how much of the shell admits a partition into independent sets, a colouring question.

Theorem 5.6. *$K(38) \ge 591\,612$.*

Proof sketch. Take $L = \Lambda_{24}$, $k = 14$ and $t = \frac{2}{3}$. For Z take the $1\,932$-point kissing configuration of $\mathbb{R}^{14}$, which admits a partition into 644 zero-sum triples. The $98\,280$ minimal lines of Λ_{24} admit a partition into 644 pairwise disjoint independent sets, so every line carries caps and the equator is empty: (4) gives $196\,560 - 2 \cdot 98\,280 + 2 \cdot 98\,280 \cdot 3 = 6 \cdot 98\,280 = 589\,680$, the ceiling $3K(24)$ of the family.

The pole set is non-empty even though Z already occupies every direction of the $1\,932$-point code. By (9) a pole needs 30° of separation from the directions; the covering radius of that code is 54.7°. There is a rotated copy of the code no point of which lies within 30° of Z, so all $1\,932$ of them are poles and the total is $589\,680 + 1\,932 = 591\,612$, which is exactly the family's ceiling $3K(24) + K(14)$.

Such a rotation is not found by sampling. A 30° cap has relative measure 1.50×10^{-5} on S^{13}, so the $1\,932$ caps cover at most 2.89% of it; a random rotation accordingly loses 56 points on average, with standard deviation 11, and an independent-points model gives e^{-56} for a clean one. It is found instead by descending on $\mathrm{SO}(14)$. Scale the w_i to $|w_i|^2 = 8$, so that the separation asks for $\langle Qw_i, w_j\rangle \le 4\sqrt{3}$; the objective $f(Q) = \sum_i \operatorname{softplus}\big(\operatorname{smax}_j \langle Qw_i, w_j\rangle - 4\sqrt{3}\big)$ is then differentiable, smax being a smooth maximum, and the geodesic step $Q \mapsto \exp(-\eta \operatorname{skew}(\nabla f\, Q^{\mathsf{T}}))Q$ keeps the iterate on the group. The real solution is then made rational by the Cayley transform, $(I-S)(I+S)^{-1}$ being rational and orthogonal for every rational skew-symmetric S, so the verification stays exact: $Q = N/D$

with N integral, $NN^{\mathsf{T}} = D^2 I$ and $|\langle Nw_i, w_j\rangle| \le \operatorname{isqrt}(48D^2)$ for all $1\,932^2$ pairs. $\square$

The partition into independent sets is the substantial step. Let $b(x,y) = \langle x,y\rangle \bmod 2$. Since L is even, b descends to an alternating bilinear form on the $\mathbb{F}_2$-space $L/2L$. For a totally isotropic subspace $R \subset L/2L$ of dimension 4 put

$$X_R = \{\, \pm u \;:\; u \in \Lambda,\; b(u,x) = 0 \text{ for every } x \in R \,\},$$

which for the subspaces used here contains 6 120 of the 98 280 lines. Every known maximum independent set of $\Gamma_\gamma(\Lambda_{24})$ lies inside such a set. Local search restricted to one X_R returns independent sets of 230 to 244 lines, whereas the same search on the whole graph returns 152. The explicit partition is in [25].

5.8. **Dimensions 49–61 and 73–95.** Above dimension 48 the same construction runs over P_{48} and Γ_{72}. Two things change. First, by Corollary 5.2, P_{48} admits only antipodal pairs while Γ_{72} admits triples; this is the one structural difference between the two ranges. Second, the construction now needs many disjoint independent sets; the gain is proportional to their total size.

Proposition 5.7. *In the one-point counts of Proposition 4.8, the graph* $\Gamma_\gamma(\Lambda_{24})$ *has* 98 280 *vertices and is regular of degree* $n[2] = 4\,600$*. The graph* $\Gamma_\gamma(P_{48})$ *has* 26 208 000 *vertices and is regular of degree* $n[3] = 36\,848$*. The graph* $\Gamma_\gamma(\Gamma_{72})$ *has* 3 109 087 800 *vertices and is regular of degree* $n[3]+n[4] = 14\,086\,968$*. By the Caro–Wei bound they contain independent sets of size at least* 22*,* 712 *and* 221*.*

Search on explicit coordinates finds independent sets of 248 lines in the first of these graphs, 7 069 in the second and 2 118 in the third. The ratio of the size found to the Caro–Wei bound is 11.3, 9.93 and 9.58.

Two ways of producing a disjoint family are used. The first takes automorphisms: if $g \in \operatorname{Aut}(L)$ and C is independent then so is Cg, and two random images of a set of size c overlap in about $c^2/\#\{\text{lines}\}$ lines, which we delete from the later image. This gives 1 400 pairwise disjoint independent sets of P_{48}, covering 8 023 414 lines or 30.6% of the minimal lines, and 32 000 of Γ_{72}, covering 47 150 230 lines. The second runs the search itself on a pool of minimal lines, removing each set from the pool as it is found; over Γ_{72} it gives 32 000 sets covering 65 570 602 lines. The two have opposite shapes. The images of one good set begin at 2 106 lines and decay to 867, while the sets found afresh lie between 1 932 and 2 118. A dimension that reads few sets is better served by the first family and one that reads many by the second, the crossover falling at dimension 86. The two are never mixed, disjointness across families not being claimed.

Theorem 5.8. *For* $49 \le n \le 61$ *and* $73 \le n \le 95$ *the lower bounds of Table 4 hold.*

Proof sketch. Apply (4) with $L = P_{48}$, $t = \frac{3}{4}$ and $|Z_i| = 2$ for $49 \le n \le 61$. For $73 \le n \le 95$ apply it with $L = \Gamma_{72}$, taking $t = \frac{2}{3}$ and $|Z_i| = 3$ where a partition of Z into zero-sum triples is available, and $t = \frac{3}{4}$ with $|Z_i| = 2$ otherwise. The direction set Z is a root system for $k \le 8$ and a cross-section of Λ_{24} for larger k; if its partition has r parts, take the r largest available independent sets. At $t = \frac{3}{4}$ the poles are unconstrained and contribute $K(k)$; at $t = \frac{2}{3}$ we take for W a rotated copy of Z restricted to the points satisfying the $30°$ separation of (9), which is between 99% and 100% of it; it is all of Z at $k = 3$, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15 and 16, and reaches $K(k)$ at $k = 3$, 5, 6, 7, 8, 9, 10, 14 and 16.

The choice of Z is not forced. Conditions (7) and (8) ask of it only that it be a $60°$ code admitting a partition into parts internally at cosine $\le -\frac{1}{2}$, so a larger such code with a perfect partition is strictly better, and at $k = 12$ and 13 the cross-section taken was not the largest available. Let π be an automorphism of the Golay code of cycle shape $1^6 3^6$, acting on Λ_{24} by permuting coordinates. Its fixed space is 12-dimensional — six fixed points and six 3-cycles — and on the orthogonal complement, also 12-dimensional, π has no eigenvalue 1, so $I + \pi + \pi^2 = 0$ there and every orbit $\{v, \pi v, \pi^2 v\}$ of a vector of that complement sums to zero. The 756 minimal vectors of Λ_{24} lying in the complement have determinant 3^6 at minimal norm 4, which is the Coxeter–Todd lattice K_{12}, and their partition into 252 zero-sum triples is the orbit set of π: a group acting, not a packing. Adjoining the one further direction carrying the most minimal vectors — 162 of them — gives a 13-dimensional section of 918, against Λ_{12}'s 648 and Λ_{13}'s 906. Thirteen is odd, so no order-3 isometry of $\mathbb{R}^{13}$ acts freely and those 306 triples are packed rather than given.

The same identity settles four partitions that a packing search had left short. An isometry S with $I + S + S^2 = 0$ has eigenvalues ω and $\bar{\omega}$ only, so it acts freely and all its orbits are zero-sum triples; an S-invariant Z is therefore partitioned perfectly by them, and those eigenvalues come in conjugate pairs, so this can happen only in even dimension. At $k = 16$, 18, 20 and 22 such an S exists and the partitions improve from 1 434 triples and 7 pairs to 1 440 triples, from 2 457 and 11 to 2 466, from 5 782 and 23 to 5 800, and from 16 594 and 49 to 16 632. At $k = 20$ the isometry condition alone leaves the first level of the search 664 ways wide and a second invariant is needed: S carries zero-sum triples to zero-sum triples, so it preserves the number of them through a vector, and filtering the candidate images by that degree narrows the level to eight. At $k = 22$ that degree is constant and the filter buys nothing; the isometry condition settles it there on its own, in twelve complete assignments. The six dimensions this moves are 84, 85, 88, 90, 92 and 94, by 549 638 spheres in total.

The odd k carry no such S, and their partitions had been left short by a greedy that never revisits a placed triple: 9, 13, 25 and 60 triples at $k = 17$, 19, 21 and 23. What closes them is that the hypergraph of zero-sum triples is *linear* — two vectors lie in at most one triple, since the third is forced to be $-(a+b)$ — which determines the move set of a local search completely. If a triangle $\{u,v,w\}$ has u uncovered and v, w in a common part, that part's third vector is $-(v+w)=u$, which is uncovered; so it does not happen. Evicting one part therefore always leaves the number of uncovered vectors unchanged, and every neutral or improving move is a pair of uncovered vectors at $120°$. Enumerating those pairs instead of sampling triangles takes the partitions from 1 773 triples and 11 pairs to 1 782, from 3 543 and 16 to 3 556, from 9 215 and 34 to 9 240, and from 30 990 and 82 to 31 050, perfect in every case, and moves dimensions 89, 91, 93 and 95 by 286 996 spheres.

The economy that suggests itself is not available. $Z=-Z$ and a triple's signs are fixed up to a global one, so its triples come in antipodal pairs and the search appears to halve — and an S-invariant partition is antipodally symmetric for the same reason, the orbit of $-v$ being the negative of the orbit of v. Summing $[a]+[b]+[c]=0$ in $\Lambda/2\Lambda$ over such a partition gives $\sum[v]=0$ over the $|Z|/2$ minimal lines, and for Λ_k that fails at $k=11$, 13, 15, 17, 19 and 23. On the vectors the same sum is doubled and says nothing, which is why the search runs there; a search on the lines stalls one triangle short at $k=17$, at 890 of 891, for arithmetic reasons rather than for want of looking.

A copy of Z cannot exceed $|Z|$, and at $k=19$, 20 and 21 the direction set is the minimal-vector set of Λ_k, which is smaller than $K(k)$: 10 668, 17 400 and 27 720 against 11 948, 19 448 and 29 768. Those three layers were short by choice of ansatz rather than by geometry, since (9) and (10) ask of W only that it be a $60°$ code lying $30°$ from Z — not that it be a copy of anything. Taking for W a rotated copy of the record configuration of $\mathbb{R}^k$ instead gives 11 934, 19 415 and 29 712 poles. At $k=19$ and 20 that configuration is available in the coordinates the direction set is already written in, Λ_k there being $4\binom{n}{2}$ vectors $(\pm2,\pm2,0,\dots,0)$ together with the 128 even-sign patterns on each octad of a shortened Golay code; taking the octad signs odd and adjoining $((-1)^{c_1},\dots,(-1)^{c_n})$ for c in the dual of the octad code is Cohn–Li's construction, and it attains $K(k)$.

At $k=21$ the direction set is written in 24 coordinates spanning 21, so a configuration built in $\mathbb{R}^{21}$ must be carried into the axis space by a rational similarity: k orthogonal vectors of a common norm μ in the span. Whether one exists is decided by the discriminant of that space, since an orthogonal basis of norms μ has determinant μ^k modulo squares. At $k=18$ and $k=22$ the discriminant is 3 and k is even, so none exists at all and no configuration built in the standard $\mathbb{R}^k$ can be placed there rationally; at $k=21$ it is 1, so μ must be a square, which is why a frame drawn from the norm-32 direction vectors is always one short and one drawn from norm-4 vectors of the span is not.

That is a statement about the standard metric and not about the configuration. Three further direction sets are cross-sections written in more coordinates than they span, at $k = 15$, 17 and 18, and there the record configuration is not rebuilt but read: Cohn's coordinate data set gives it as exact integers together with an integer metric $\operatorname{diag}(c)$, which for $k = 15$, 17, 18 is $\operatorname{diag}(1^{14}, 2)$, $\operatorname{diag}(1^{16}, 2)$ and $\operatorname{diag}(1^{16}, 2, 6)$. A frame then has to satisfy $M^{\mathsf{T}}M = \mu \operatorname{diag}(c)$ and the discriminant condition becomes $\mu^k \det \operatorname{diag}(c) \equiv$ disc. At $k = 18$ that is $12 \equiv 3$, the discriminant of the axis space, and a frame does exist: sixteen columns $v_1 \pm v_2, \ldots, v_{15} \pm v_{16}$ of norm 64 built from mutually orthogonal direction vectors, $2v_{17}$ of norm 128, and for the last one the orthogonal complement of those seventeen inside the span, which is a line and whose norm class is therefore forced — it comes out at 6 modulo squares, which is what the frame needs. The rotation is taken to be a product of rational plane rotations of $\operatorname{diag}(c)$, each of denominator D with $a^2 + (c_j/c_i)b^2 = D^2$, since a full-rank Cayley transform on a space of this rank has denominator about d^k and cannot be rounded finely enough to spend the margin. The layers go from 2 340, 5 338 and 7 374 to 2 562, 5 704 and 7 624 poles, against $K(k) = 2\,564$, 5 730 and 7 654.

Dimension 83 is no longer an exception either. Its configuration is realised over $\mathbb{Z}[\sqrt{2}]$, but only the configuration is: a rational $Q = N/D$ with $Q^{\mathsf{T}}CQ = C$ carries the directions to poles in the same ring, and (9) becomes $(p + q\sqrt{2})^2 \le 972D^2$ for integers p, q, decided exactly. All $604 = K(11)$ of them clear. $\square$

Every dimension from 81 to 95 now carries its own construction: dimension 81 exceeds dimension 80 by 185 042, so no entry of Table 4 in this range rests on monotonicity of K. The construction also runs in dimensions 62 and 63, where it reaches 64 217 822 and 67 365 752 and is beaten; §6.2 treats those two.

The layered construction stops winning at 96. Throughout $n \le 95$ the Edel–Rains–Sloane total of §6 stays below Γ_{72}. Its largest term is $A(n_0, \lceil n_0/4 \rceil)$, for which the code tables give at most 2^{30} up to $n_0 = 92$, or 17% of 6 218 175 600. At $n_0 = 93, 94, 95$ they give 2^{31}, 2^{31} and 2^{32}, and the whole chain evaluated at those three lengths gives 2.68×10^9, 2.68×10^9 and 4.83×10^9. At $n_0 = 96$ the code $[96, 33, 24]$ makes the leading term $A(96, 24) \ge 2^{33}$, at which the chain $(96, 24, 6, 1)$ totals 12 886 999 232, against the 6 480 558 568 the layered construction reaches there. Dimension 96 is therefore Edel–Rains–Sloane's rather than Γ_{72}'s, and §6.3 takes it on that footing — as §6.2 does dimensions 62 and 63.

Every input to those evaluations is a construction, so each is a lower bound for the Edel–Rains–Sloane count. A lower bound above Γ_{72} settles which of the two constructions wins there; a lower bound below it settles nothing by itself. §8.2 measures the margin in the dimensions above 48.

And the table stops there. K is superadditive: two 60° codes placed on orthogonal subspaces meet at inner product 0, so $K(96 + m) \ge K(96) + K(m)$, and every dimension above 96 already follows from the row for 96 together with a published low-dimensional record. A row for dimension 100 reading $12\,886\,999\,232+24$ would be arithmetic on the row above it. What that leaves open is whether anything overtakes the implied floor above 96, as Edel–Rains–Sloane overtakes Γ_{72} *at* 96. The layered construction does not: it ends at $k=24$. The chain does, but by a margin that decides nothing — evaluated through $n=128$ from the same tables used here, it exceeds the implied floor by at most 0.016%, at $n=114$. So 96 is where the table stops saying anything, which is why it stops there.

6. The Edel–Rains–Sloane chain

Four improvements are code-theoretic. The Edel–Rains–Sloane construction [18] builds a 60° code from a chain $n \geq n_0 \geq 4n_1 \geq 16n_2 \geq \cdots$ of support sizes, giving

$$K(n) \;\geq\; \sum_{\nu} A(n, n_\nu, n_\nu)\, A\big(n_\nu, \lceil n_\nu/4 \rceil\big).$$

The construction is defined for every n, but its authors evaluate it only at $n = 32, 36, 40, 44, 64, 80, 128$. Elsewhere its value is a question about the current code tables rather than about the construction.

6.1. **Dimension 39.** In dimensions 38 and 39 the chain that maximises the total has three levels, $(n, 8, 2)$, and reads

$$K(n) \;\geq\; A\big(n, \lceil n/4 \rceil\big) + 128\, A(n, 8, 8) + 4\binom{n}{2}.$$

Theorem 6.1. *$K(39) \geq 756\,116$.*

Proof. Echols [17] improves $A(39,8,8)$ from $3\,323$ to $3\,324$; Brouwer's table [4] carries the new value. With $A(39,10) \geq 327\,680$ [44],

$$327\,680 + 128 \cdot 3\,324 + 4 \cdot 741 \;=\; 327\,680 + 425\,472 + 2\,964 \;=\; 756\,116. \quad \square$$

The gain is $128\,(3324-3323) = 128$. Echols [17] computes kissing numbers from the fixed level-one length $n_0 = 32$, which in dimension 39 falls below the record; that paper therefore reports no value there. The chain optimises at $n_0 = n$ in dimensions 38, 39, 40, 43 and 44; among these, 38 and 39 are the two whose totals exceed the published entry; §5.7 supersedes dimension 38. That entry is 755 988, which [6] attributes to the Edel–Rains–Sloane construction; of the three dimensions named in [40] it is dimension 43, not dimension 39, whose entry that table credits to the antipode construction. That paper also reconstructs the twelve cross-section packings of 1982 in dimensions 38–47 and computes their kissing numbers, reporting that they exceed the best previously tabulated lower bounds in dimensions 42–47. Dimensions 38 and 39 are the only two this paper claims inside that span of 38–47, and neither lies in 42–47.

The verification in [25] checks the constant-weight code over all $\binom{3324}{2}$ pairs and the ten geometric conditions of the chain as exact rational inequalities, three of them tight. The value $A(39,10) \geq 327\,680$ is taken from

[44] and is not verified here; the existing record 755 988 rests on the same value.

6.2. **Dimensions 62 and 63.** The first term of the chain needs no chain. A binary code of length n_0, dimension m and minimum distance d sends each codeword c to the unit vector

$$x(c) \;=\; n_0^{-1/2}\big((-1)^{c_1},\dots,(-1)^{c_{n_0}}\big),$$

and two codewords at Hamming distance j satisfy $\langle x(c), x(c')\rangle = (n_0 - 2j)/n_0$. Every pair is therefore at 60° or more exactly when $d \geq \lceil n_0/4\rceil$; the 2^m sign vectors are then a kissing configuration of $\mathbb{R}^{n_0}$, hence of $\mathbb{R}^n$ for every $n \geq n_0$:

$$K(n) \;\geq\; \max_{n_0 \leq n} A\big(n_0, \lceil n_0/4\rceil\big).$$

Since $\lceil 62/4\rceil = \lceil 63/4\rceil = 16$, the codes $[62, 26, 16]$ and $[63, 27, 16]$ of [21] qualify, giving 2^{26} and 2^{27} sign vectors. Both are shortenings of one $[64, 28, 16]$; shortening an $[n, m, d]$ code gives $[n-1, m-1, \geq d]$. That alone already beats the values of §1.1.

The rest of the chain is not negligible: it carries Edel–Rains–Sloane's own dimension-64 total from $2^{28} = 268\,435\,456$ to $331\,737\,984$. Since $4 \cdot 16 = 64 > 63$, the largest level-one weight the chain allows in either dimension is $n_1 = 15$, and $15 \cdot 4 = 60 \leq 62$, so the chain is $(n, 15, 2)$ and

$$K(n) \;\geq\; A\big(n, \lceil n/4\rceil\big) + A(15,4)\,A(n,15,15) + 4\tbinom{n}{2}.$$

Theorem 6.2. *$K(62) \geq 71\,310\,732$ and $K(63) \geq 138\,419\,844$.*

Proof. Level zero gives 2^{26} and 2^{27} as above. For level one, index the cells of a 15×4 grid by 60 of the coordinates and let a support take one cell in each row; two supports then meet in exactly $15 - d_H$ of their choice words, so a quaternary code of length 15 and minimum distance 8 yields supports meeting in at most $7 = \lfloor 15/2\rfloor$. The cyclic $[15, 6, 8]_4$ with defining set $\{0, 1, 2, 3, 4, 5, 8, 10, 12\}$, a union of cyclotomic cosets modulo 15 under $x \mapsto 4x$, gives $A(n, 15, 15) \geq 4^6 = 4\,096$, and $A(15, 4) = 1\,024$ from the extended Hamming $[16, 11, 4]$ shortened once. Level two takes every pair of coordinates with all four sign patterns. Hence

$$67\,108\,864 + 4\,194\,304 + 7\,564 = 71\,310\,732,$$
$$134\,217\,728 + 4\,194\,304 + 7\,812 = 138\,419\,844. \qquad \square$$

Everything above level zero is built rather than cited: [25] enumerates all 4^6 codewords of the quaternary code to establish its minimum distance, checks all $\binom{4096}{2}$ support overlaps directly, enumerates all 2^{10} codewords of the $[15, 10, 4]$, and runs the whole chain end to end in coordinates at the two dimensions where it happens to reproduce the exact kissing number — $n = 8$, chain $(8, 2)$, giving the 240 points of the E_8 root system, and $n = 16$, chain $(16, 4, 1)$, giving 4 320, the Barnes–Wall value and the record — with every pair of both configurations checked. Only the two level-zero code parameters are taken from [21].

Against the values of §1.1 these are factors 1.36 and 2.64. Level zero alone already exceeds the *ceiling* of the construction of §5.8 in these two dimensions, so no family of independent sets can recover them there. Over P_{48} that construction gives $52\,416\,000 + 2\sum_i |C_i| + K(k)$ by (4) with $|Z_i| = 2$. The number of independent sets is at most the number $\lfloor K(k)/2 \rfloor$ of antipodal pairs available among the directions, and each $|C_i|$ is at most the largest independent set known, of 7 069 lines. At $k = 14$ and $k = 15$ that gives 66 075 240 and 70 543 480, short of 2^{26} and 2^{27} by 1 033 624 and 63 674 248. Closing those gaps would need independent sets of 7 605 and 31 903 lines against the largest known 7 069, or more parts than the record in dimension 14 or 15 allows. The chain is not a refinement of §5.8 in these dimensions but a different and better construction.

6.3. **Dimension 96.** At $n = 96$ the chain closes up: $96 \geq 96 \geq 4 \cdot 24 \geq 16 \cdot 6 \geq 64 \cdot 1$, so $(96, 24, 6, 1)$ is admissible, and every inequality but the last is an equality. Level zero is again a single sign code, and here it carries two thirds of the total: $\lceil 96/4 \rceil = 24$, and the code $[96, 33, 24]$ of [21] gives $A(96, 24) \geq 2^{33}$.

The two middle levels apply the grid map of §6.2 at two different factorisations of 96. Writing $96 = 24 \cdot 4$ and indexing the cells of a 24×4 grid by the coordinates, a support taking one cell in each row has weight 24, and two supports meet in exactly $24 - d_H$ of their choice words; a quaternary code of length 24 and minimum distance 12 therefore gives supports meeting in at most $12 = \lfloor 24/2 \rfloor$, and $[24, 9, 12]_4$ of [21] gives $A(96, 24, 24) \geq 4^9$. Writing instead $96 = 6 \cdot 16$, the same argument over $\mathbb{F}_{16}$ with the Reed–Solomon code $[6, 4, 3]_{16}$ gives $A(96, 6, 6) \geq 16^4$.

Theorem 6.3. *$K(96) \geq 12\,886\,999\,232$.*

Proof. The four levels contribute

$$A(96, 24) + A(96, 24, 24)\, A(24, 6) + A(96, 6, 6)\, A(6, 2) + A(96, 1, 1)\, A(1, 1),$$

in which $A(6, 2) = 32$, $A(1, 1) = 2$ and $A(96, 1, 1) = 96$ are exact, while [4] brackets $A(24, 6)$ between 16 384 and 24 106; a larger value would only raise the total, so the smaller end is substituted. With that and the three lower bounds above,

$$8\,589\,934\,592 + 262\,144 \cdot 16\,384 + 65\,536 \cdot 32 + 96 \cdot 2 \;=\; 12\,886\,999\,232. \qquad \square$$

The value of §1.1 here is the direct sum $\Gamma_{72} \oplus \Lambda_{24}$, at $6\,218\,175\,600 + 196\,560$, so the factor is 2.07. As in §6.2 the mathematics is Edel–Rains–Sloane's, and as there we find no record of their construction having been evaluated at this length, which is not among their seven worked examples; their own first remark licenses substituting any available lower bounds for the code sizes, which is all that is done above. The total is a function of those tables: with the three levels below weight 96 as constructed here — the weight-24, weight-6 and weight-1 terms, together 4 297 064 640 — the chain passes the direct sum as soon as $A(96, 24) \geq 2^{31}$, and $[96, 33, 24]$ gives 2^{33}. Of the three code inputs only the smallest is built here: [25] constructs $[6,4,3]_{16}$ and establishes its minimum distance by evaluating all 16^4 polynomials. The sign code $[96,33,24]$, which carries two thirds of the total, and the quaternary code $[24,9,12]_4$ are taken from [21].

7. The results

Table 4: All forty-eight improvements. "Previously" is [6] where that table has an entry and otherwise the direct sum or inherited value of §1.1. Dimensions 44 to 47 are absent: [40] gives larger values in dimensions 44 and 45; dimensions 46 and 47 are already in the literature (§1.4).

n	previously	this work	construction
25	197 056	197 569	layers over Λ_{24}, lens heads (§5.4)
26	198 550	199 632	layers over Λ_{24}, one triangle (§5.6)
27	200 044	201 010	layers over Λ_{24}, two coset triangles
38	566 652	591 612	layers over Λ_{24}, $k = 14$
39	755 988	756 116	constant-weight codes
49	52 416 002	52 430 140	layers over P_{48}
50	52 416 006	52 458 418	layers over P_{48}
51	52 416 012	52 500 816	layers over P_{48}
52	52 416 024	52 585 516	layers over P_{48}
53	52 416 040	52 698 222	layers over P_{48}
54	52 416 072	52 922 906	layers over P_{48}
55	52 416 126	53 299 730	layers over P_{48}
56	52 416 240	54 085 392	layers over P_{48}
57	52 416 306	54 534 176	layers over P_{48}
58	52 416 510	55 893 422	layers over P_{48}
59	52 416 604	56 505 668	layers over P_{48}
60	52 416 841	57 477 123	layers over P_{48}
61	52 417 154	59 898 694	layers over P_{48}
62	52 417 932	71 310 732	ERS chain $(62, 15, 2)$
63	52 418 564	138 419 844	ERS chain $(63, 15, 2)$
68	331 737 984	361 275 480	cross-section of Γ_{72}, $k = 4$
69	331 737 984	627 822 180	cross-section of Γ_{72}, $k = 3$
70	331 737 984	1 249 778 250	cross-section of Γ_{72}, $k = 2$
71	331 737 984	2 603 658 750	cross-section of Γ_{72}, $k = 1$
73	6 218 175 602	6 218 179 814	layers over Γ_{72}
74	6 218 175 606	6 218 192 454	layers over Γ_{72}
75	6 218 175 612	6 218 209 308	layers over Γ_{72}
76	6 218 175 624	6 218 243 016	layers over Γ_{72}
77	6 218 175 640	6 218 285 152	layers over Γ_{72}
78	6 218 175 672	6 218 377 840	layers over Γ_{72}

n	previously	this work	construction
79	6 218 175 726	6 218 529 454	layers over Γ_{72}
80	6 218 175 840	6 218 849 228	layers over Γ_{72}
81	6 218 175 906	6 219 034 270	layers over Γ_{72}
82	6 218 176 110	6 219 605 578	layers over Γ_{72}
83	6 218 176 204	6 219 621 690	layers over Γ_{72}
84	6 218 176 441	6 220 293 304	layers over Γ_{72}
85	6 218 176 754	6 220 745 414	layers over Γ_{72}
86	6 218 177 532	6 223 567 844	layers over Γ_{72}
87	6 218 178 164	6 224 703 598	layers over Γ_{72}
88	6 218 179 920	6 230 206 492	layers over Γ_{72}
89	6 218 181 330	6 233 054 848	layers over Γ_{72}
90	6 218 183 254	6 238 744 788	layers over Γ_{72}
91	6 218 187 548	6 247 801 034	layers over Γ_{72}
92	6 218 195 048	6 266 405 667	layers over Γ_{72}
93	6 218 205 368	6 294 850 360	layers over Γ_{72}
94	6 218 225 496	6 355 739 600	layers over Γ_{72}
95	6 218 268 750	6 473 065 046	layers over Γ_{72}
96	6 218 372 160	12 886 999 232	ERS chain $(96, 24, 6, 1)$

8. Remarks

8.1. **Obstructions.** Several constructions reach their exact ceiling. Details and certificates are in [25].

The Cohn–Li construction is exhausted in dimensions 18*,* 20 *and* 21*.* It adjoins to a base configuration a set of vectors indexed by words orthogonal to a constant-weight code, two such vectors being compatible exactly when their Hamming distance is at least $n/4$ [12]. The words in question form a punctured Golay code of 4 096 elements. The number adjoined is therefore the independence number of a Cayley graph on $\mathbb{F}_2^{12}$, whose connection set Σ consists of the non-zero words of weight less than $n/4$. For an abelian Cayley graph the Delsarte bound equals the Lovász theta function. In dimensions 20 and 21 both equal 2 048, which [12] attains. In dimension 20 the set Σ has five elements and they are linearly independent, so each of the 128 components of the graph is the 5-cube, bipartite with independence number 16, whence $128 \times 16 = 2\,048$. In dimension 18 the number is 768 and is attained, but the construction then totals $6\,500 + 768 = 7\,268$, short of the 7 654 that [12] prove in that dimension by a configuration of a different shape.

Dimension 19 *is closed at the value of Ho.* Here the graph has four components. Each is the Cayley graph on $\mathbb{F}_2^{10}$ with a connection set Σ of 21 elements; the construction produces $10\,668 + 4\alpha$ points, where α is the independence number of one component. Exactly one element t of Σ has odd weight. Writing H for the even-weight hyperplane, the two halves H

and $H + t$ induce the same 512-vertex graph G_0, and the edges between them form the perfect matching $x \mapsto x + t$; hence $\alpha \le 2\alpha(G_0)$. Constraint programming, with the inequality that an independent set meets a 5-cycle of G_0 in at most two vertices, proves $\alpha(G_0) = 160$ optimal. Hence $\alpha \le 320$, which [22] attains; 11 948 is the exact ceiling of the construction.

For a kissing configuration $X \subset S^{d-1}$ put

$$m(X) = \min_{z \in S^{d-1}} \max_{x \in X} \langle z, x \rangle.$$

A further point may be added to X exactly when $m(X) \le \frac{1}{2}$, so X is maximal exactly when $m(X) > \frac{1}{2}$; we call $2m(X) - 1$ the *margin* of X. The values of m below are obtained by numerical maximisation of $|z|$ over the polytope $\{z : \langle z, x \rangle \le \frac{1}{2}$ for all $x \in X\}$, whose maximum is $1/(2m(X))$; where an exact algebraic value is quoted, it is confirmed in rational arithmetic.

Λ_{21}, Λ_{22} and Λ_{23} are maximal spherical codes, with $m = \sqrt{8/29}$, $\sqrt{3/11}$ and $\sqrt{4/15}$, so no point may be added to any of them; the level argument of §5.4 has no analogue there. The remaining cells of §4.6 add nothing either. Fix $u \in \Lambda$ and project the shell of Λ_{24} onto $u^{\perp}$. The cell $n[0]$ already lies in $u^{\perp}$ and is the minimal shell of Λ_{23}. Two projected vectors taken from opposite signs of the cell $n[1]$, or of the cell $n[2]$, can have inner product above $\frac{1}{2}$; call such a pair a conflict. At most one sign of each cell is therefore available. On the union of $n[0]$ with one sign of each of the other two, conflicts occur only between $n[0]$ and the rest: each vector of $n[0]$ conflicts with 1 024 of the 47 104 and with 44 of the 4 600, while these conflict with 2 025 and 891 vectors of $n[0]$ respectively. Weighting each edge by 1/2025 or by 1/891 accordingly gives a fractional matching under which every vector of $n[0]$ carries load $1024/2025 + 44/891 < 1$. The bipartite matching polytope is integral, so a matching saturates the 51 704 vectors outside $n[0]$. By König's theorem the largest kissing configuration obtainable this way has exactly 93 150 points.

Every published record in dimensions 9–19 *is maximal,* its largest cosine m exceeding $\frac{1}{2}$ by between 4.4% (dimensions 17 and 19) and 34.2% (dimension 10). Dimensions 14 and 15 have the same value $m = \sqrt{2/7}$.

The antipode construction is bounded by a spherical cap. Fix linearly independent $u_1, \ldots, u_k \in \Lambda$, let π denote the orthogonal projection of $\mathbb{R}^n$ onto their span, and for $p \in \pi(L)$ put $f(p) = \#\{w \in \Lambda : \pi(w) = p\}$, so that $f(0) = |S|$ in the notation of Lemma 4.1. The construction of [40] chooses a finite $T \subset \pi(L)$ whose largest squared pairwise distance δ is less than μ and returns the maximum over $p \in T$ of $\sum f(q - p)$, the sum running over the $q \in T$ with $|q - p|^2 = \delta$. Over all such T this maximum equals the largest weight of a clique, taken over all $\delta \in (0, \mu)$, in the graph on $\{v \in \pi(L) : |v|^2 = \delta,\ f(v) > 0\}$ in which v carries weight $f(v)$ and v, v' are adjacent when $\langle v, v' \rangle \ge \delta/2$. Adjacency means an angle of at most 60°, so a clique lies in a spherical cap of that angular radius about any of its members, and the construction cannot exceed the total weight of the heaviest such cap.

Two members lie within 30° of their bisector, but the radius does not stay there: three members pairwise at exactly 60° need $\arccos\sqrt{2/3} = 35.26°$. The cap bound improves on $f(0)$ alone only when $f(0)$ is already within about 10% of the value it has to exceed, as in dimensions 44 through 47.

8.2. **The margin above dimension 48.** Among the dimensions at issue the tables have a single entry, at $n = 80$, where it agrees with the direct sum of §1.1. Everywhere else above dimension 48 the value a bound has to exceed is not a tabulated one but the largest a known construction gives. That construction is Edel–Rains–Sloane, evaluated in the dimension itself rather than inherited from a smaller one (Remark 4.13). We therefore ran the chain of §6 in every dimension of Table 4 above 48, with the best published code at each level, and asked of each bound how large the level-one constant-weight code $A(n, n_1, n_1)$ would have to be for the chain to overtake it. Divided by the best code available at those parameters, that is the factor by which the published constructions would have to be beaten.

No bound of Table 4 falls below the chain, and the factors are far from uniform. *Dimension* 68 *is the most exposed bound in this paper*: its margin over the chain is 8.9%, its level-one weight is 16, and 16 is the one weight at which Edel–Rains–Sloane built a strong code of their own, giving $A(64, 16, 16) \geq 30\,828$. A code with $A(68, 16, 16) \geq 45\,235$ would overturn Theorem 4.12 in that dimension, a factor 1.47; for calibration, 30 828 beats the grid map of §6.2 at the same parameters by 1.88. No $A(68, 16, 16)$ has been published, so the exposure is to a better code and not to the literature as it stands. The next smallest factor is 4.06, in dimension 95; every other dimension exceeds 5. Dimensions 62, 63 and 96 are the chain, so the question does not arise there.

8.3. **The maximum independent set in $\Gamma_\gamma(\Lambda_{24})$.** Section 5 depends on the largest number of minimal lines of Λ_{24} that are pairwise at $|\cos| \leq \frac{1}{4}$. The record is 248 lines [29], after 240 [9] and 244 [23]; the best upper bound is $\lfloor 4680/11 \rfloor = 425$ [9, §7]. By (4) one further line in every independent set contributes $2\sum_i(|Z_i| - 1)$, so raising the record by a single line gives +2, +8, +16, +32, +52, +96 and +168 in dimensions $25, \ldots, 31$ respectively.

The Delsarte linear programme, the Lovász theta function and the Hoffman ratio bound on the association scheme whose relations on the 98 280 lines are $|\langle u, u'\rangle| \in \{0, 1, 2\}$ all give 4680/11, so no two-point relaxation improves it. The three-point semidefinite bound of Bachoc and Vallentin [1] does not improve it at any truncation computed, the positivity constraints never being active. Closing the gap requires a four-point relaxation or an argument using the lattice structure.

8.4. **Dimension 17.** The layered family over a Λ_{16} equator reduces in dimension 17 to a single integer: its largest member has $5\,346 + 2\alpha$ points, where 5 346 is the kissing number of Λ_{17} and α is again an independence number in a Cayley graph. A Parseval argument excludes the support sizes

7 and 8, leaving the supports of [12]; the remainder of the configuration is exactly 512 vectors up to sign.

The value $\alpha = 192$ is attained by [12], and it is optimal. The vertex set is not an arbitrary 1 024-element set: it is $K' = \{b \in \mathrm{RM}(2,4) : |b \cap S_0| \text{ even}\}$, a binary linear code, hence an elementary abelian group $\mathbb{F}_2^{10}$, and $W_4 \subseteq K'$ is symmetric. The graph is therefore a translation scheme, Delsarte's linear programming bound applies, and for a Cayley graph on an abelian group that bound is exactly ϑ'. Its value is 192.

Proposition 8.1. $\alpha = 192$*, so the layered family of dimension* 17 *contains no configuration with more than* 5 730 *points.*

Proof. Let $C \subseteq K'$ have no two elements differing by a member of W_4. For any $g \colon K' \to \mathbb{Q}$ with $\widehat{g} \geq 0$,

$$0 \;\leq\; \tfrac{1}{1024} \sum_{y} \widehat{g}(y) \Big| \sum_{c \in C} \chi_y(c) \Big|^2 \;=\; \sum_{c,c' \in C} g(c - c') \;\leq\; |C|\, g(0) - |C|\big(|C| - 1\big)$$

whenever $g \leq -1$ off $W_4 \cup \{0\}$, so $|C| \leq 1 + g(0)$. Take g whose normalised Fourier coefficients $b_y = \widehat{g}(y)/1024$ depend only on the eigenvalue $\lambda(y) = \sum_{w \in W_4} \chi_y(w)$ and on whether y is trivial on the fibre subgroup. With $b_y = 0$ on the remaining 118 characters:

$(\lambda,\ y\vert_H$ trivial$)$	$(30, \text{no})$	$(12, \text{yes})$	$(12, \text{no})$	$(6, \text{no})$	$(-4, \text{no})$	$(-2, \text{no})$
$\#\{y\}$	16	10	120	160	360	240
b_y	65/80	40/80	30/80	28/80	10/80	9/80
mass	13	5	45	56	45	27

Then $b \geq 0$ and $g(0) = \sum_y b_y = 191$, while $g \leq -1$ at all 963 points outside $W_4 \cup \{0\}$: exactly -1 at 887 of them, $-\frac{21}{5}$ at 60 and -9 at 16. Scaled by 80 every quantity is an integer, so the verification is one Walsh–Hadamard transform in exact arithmetic. This gives $\alpha \leq 192$; the configuration of [12] is an independent set of size 192 — six pairwise non-adjacent fibres — so $\alpha = 192$. ☐

Two coarser relaxations both stop at 256, and for the same reason. The spectrum of the graph is $60^1 30^{16} 12^{130} 6^{160} (-2)^{240} (-4)^{375} (-10)^{96} (-20)^6$, so the ratio bound is $1024 \cdot 20/80 = 256$; and the quotient on the 32 fibres has clique number 4, forcing a cover by eight 4-cliques of exact local bound 32. Both are satisfied by the uniform fractional point 1/4 on the fibres, as is every clique, block-pair and odd-hole cut, so no relaxation phrased on the quotient can beat 256. The linear programme over the group does not factor through the quotient. As above, the conclusion applies to the family alone.

8.5. **Verification.** The package [25] produces and checks every number above. It applies the following standards.

- *No floating-point arithmetic decides any bound of Table 4.* Every linear programming bound carries an exact rational dual certificate (Lemma 4.6); the solver is used only to propose a dual vector. What decides the remaining bounds is an integer comparison. Two scans are too large to run in arbitrary-precision integers — the axis layer of dimension 95 compares $93\,150^2$ inner products — and are run in floating point on integer operands bounded so that every product and every partial sum is exactly representable. The bound is asserted before the scan; where it does not hold, the operand is split into two limbs at bit 31 and recombined exactly.

- *Every Gram matrix is proved realizable before use*, by Lemma 4.7 or by exhibiting a k-tuple that carries it. The ones neither licensed nor exhibited are recorded and discarded.
- *Configurations small enough to write down are checked in coordinates.* The dimension-27 configuration is verified by two deliberately different arguments, one of which compares all $\binom{200540}{2} = 20\,108\,045\,530$ pairs; that configuration is now superseded (§5.6), and the two that replace it are verified in exact arithmetic, the head–axis inequalities in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.
- *The cross-section method is validated against every independently known value* (Table 2). The inequalities and the counting formula of the layered construction are re-derived from first principles in exact rational arithmetic, independently of the code that applies them.
- *Large intermediate data is regenerated by deterministic scripts* that re-verify every property the claims depend on.

Use of AI assistance

The construction of §5.5 was found by Claude Opus 5 (Anthropic, maximum reasoning effort) in a chat session in which the author provided the problem, Cohn's table [6] and Cohn's data set [7]. It appeared at the fortieth prompt; most of the thirty-nine earlier prompts were some form of "try harder".

The remaining constructions, the package [25] and this exposition were then developed by the author with Claude Code (Anthropic), a coding agent built on the same model; those of §5.4 and §5.6 with Claude Fable 5.1 in the same way, inside the joint work with H. Cohn and B. Lindow. The agent proposed constructions, implemented and ran the verification code, and drafted the exposition. The author set the direction, selected the lines to pursue and reviewed the text and the certificates.

In neither setting did the model predict success, and it said so in plain terms: "I have not produced a new SOTA ... continuing would be me generating variations faster than either of us can check them."

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